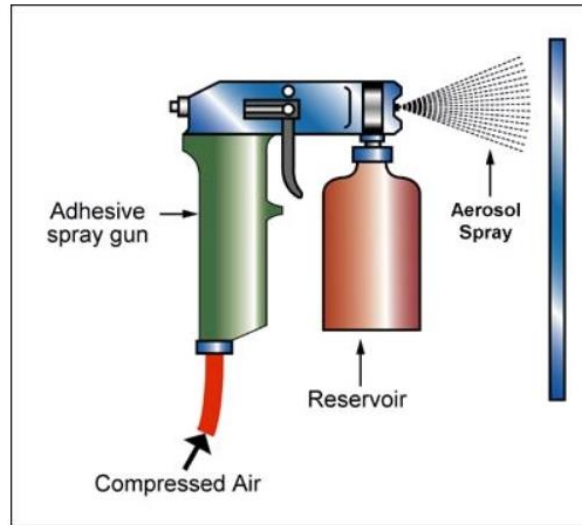


Joining adhesive – flexible adhesive

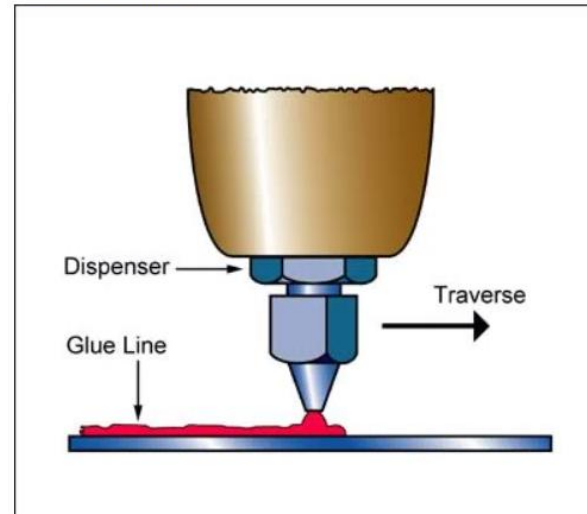
Adhesive spray gun, aerosol spray, reservoir, compressed air



Joining adhesive – rigid adhesive

Dispenser, glue line, traverse

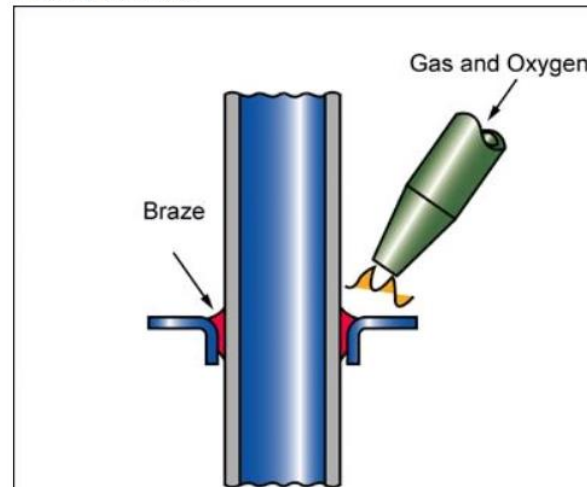
Process schematic

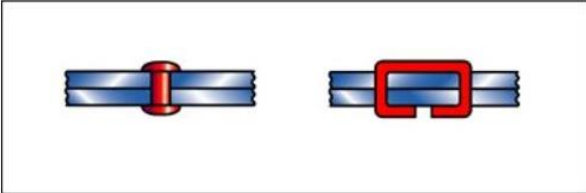
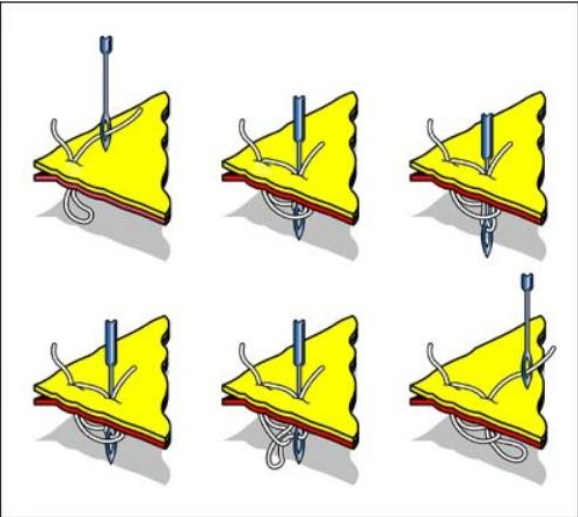
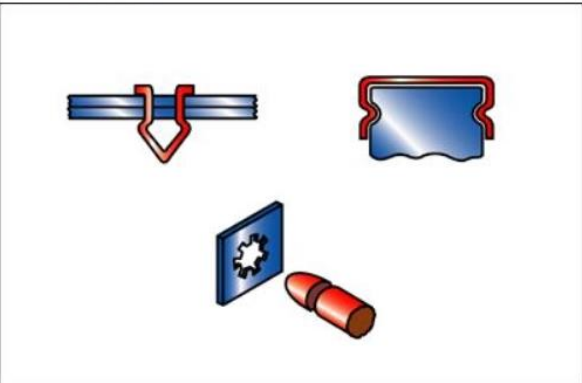
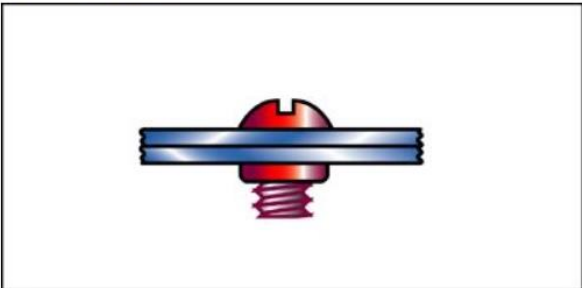


Brazing / Soldering

Braze, gas and oxygen

Process schematic

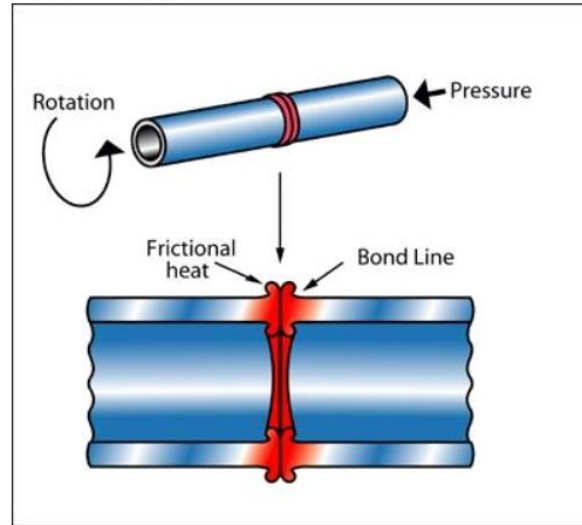


Rivets and staples C shape, red circle	Process schematic 
Sewing Needle, thread, yellow triangle,	Process schematic 
Snap fit Red crayon looking mf, red spear tip, triangle, 7 side hole, starfish	Process schematic 
Threaded fasteners Screw Bolting	Process schematic 

Mechanical Welding – Friction Welding

Rotation, pressure, metal tube, thin pipe, thin rod, blunted joined ends, frictional heat, bond line

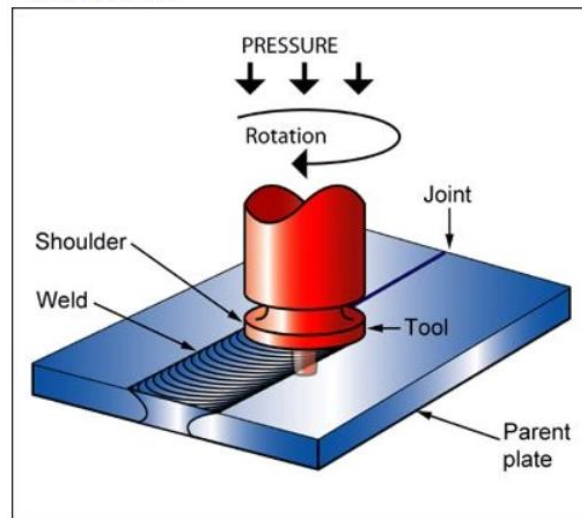
Process schematic



Mechanical Welding – Friction stir weld

Weld, shoulder, rotation, pressure, tool, parent plate, joint

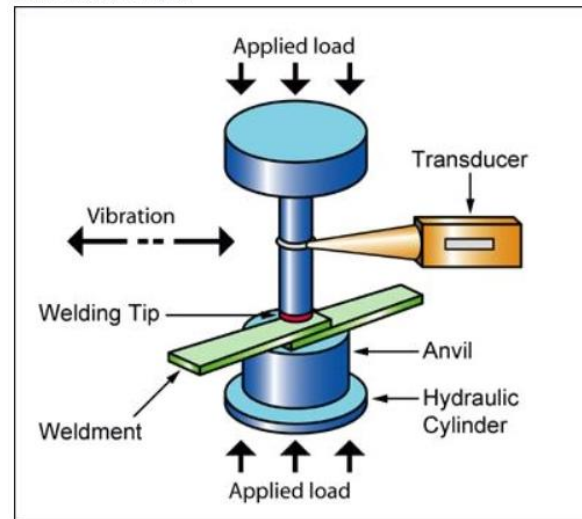
Process schematic



Mechanical Welding – Ultrasonic welding

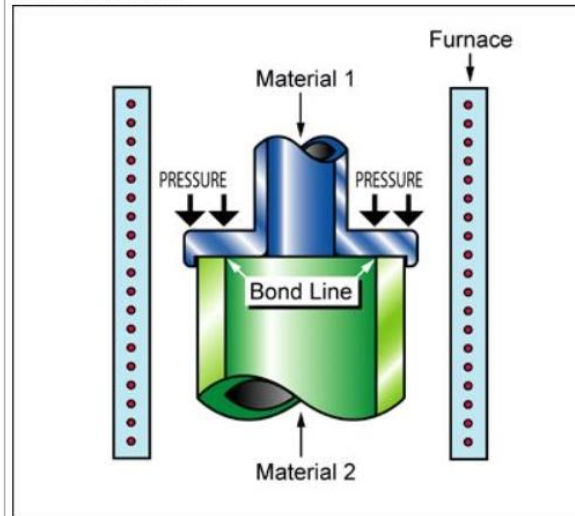
Vibration, applied load, transducer, welding tip, anvil, weldment, hydraulic cylinder, applied load

Process schematic



Thermal welding -
Ceramics/metals -
Diffusion and glaze
bonding

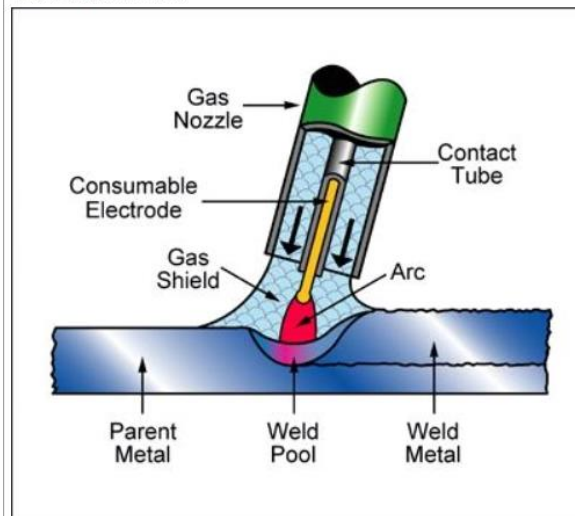
Process schematic



Fusion welding –
metals – high
temperature – Gas
metal arc welding

Gas nozzle,
consumable
electrode, gas shield,
contact tube, arc,
parent metal, weld
pool, weld metal
Vacuum cleaner
looking

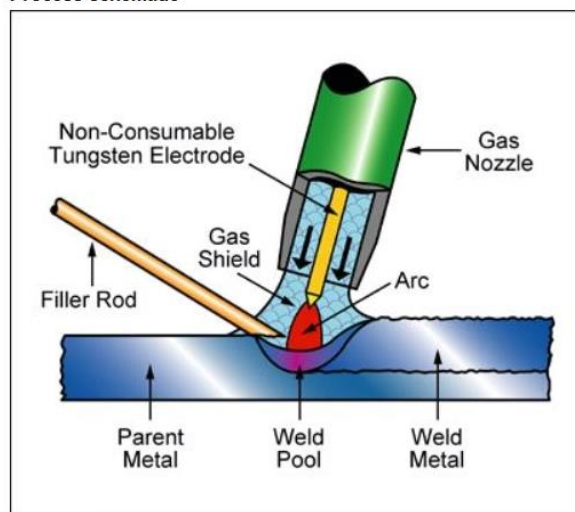
Process schematic



Thermal welding –
metals – high
temperature – Gas
tungsten arc

Non-consumable
tungsten electrode,
gas nozzle, gas shield,
filler rod, arc, parent
metal, weld pool, weld
metal

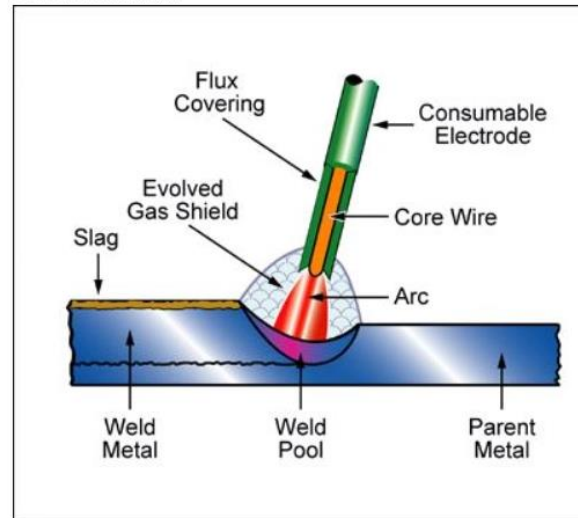
Process schematic



Thermal welding –
Manual Metal Arc
(MMA)

Flux covering,
consumable
electrode, evolved gas
shield, core wire, slag,
arc, weld metal, weld
pool parent metal

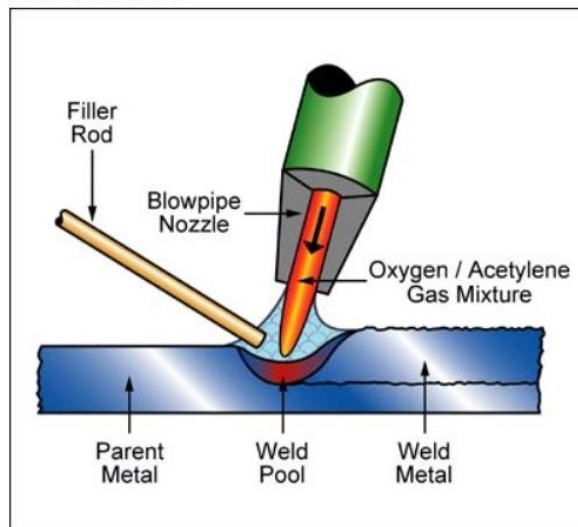
Process schematic



Thermal welding –
metals – Oxyacetylene
welding

Filler rod, blowpipe
nozzle,
oxygen/acetylene gas
mixture, parent metal,
weld pool, weld metal

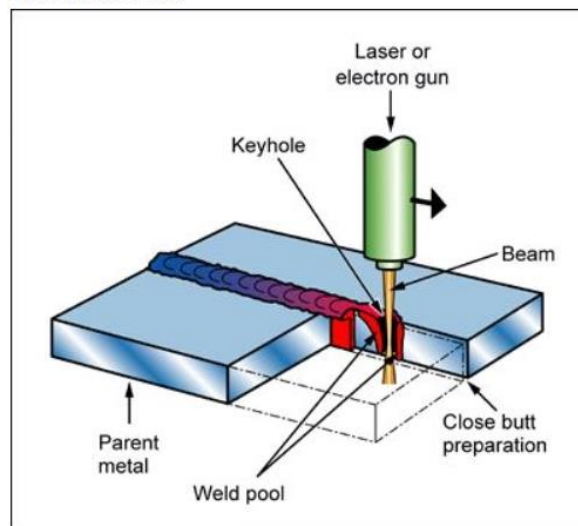
Process schematic



Thermal welding –
metals – high
temperature – power
beam
Electron beam
welding

Laser or electron gun,
keyhole, beam, weld
pool, parent metal,
close butt preparation

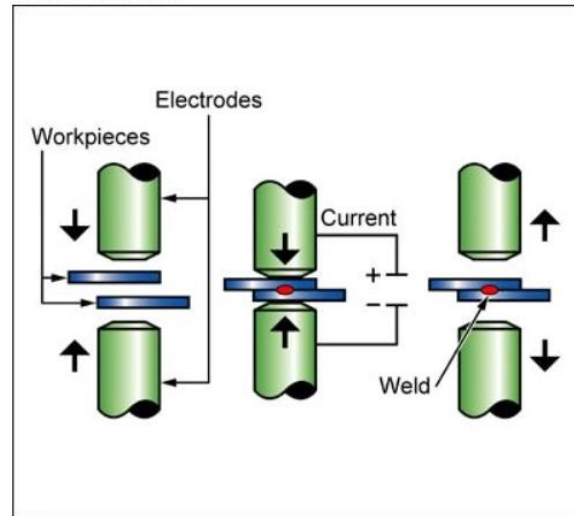
Process schematic



Thermal welding –
metals – high
temperature –
resistance welding

Workpieces,
electrodes, current,
weld

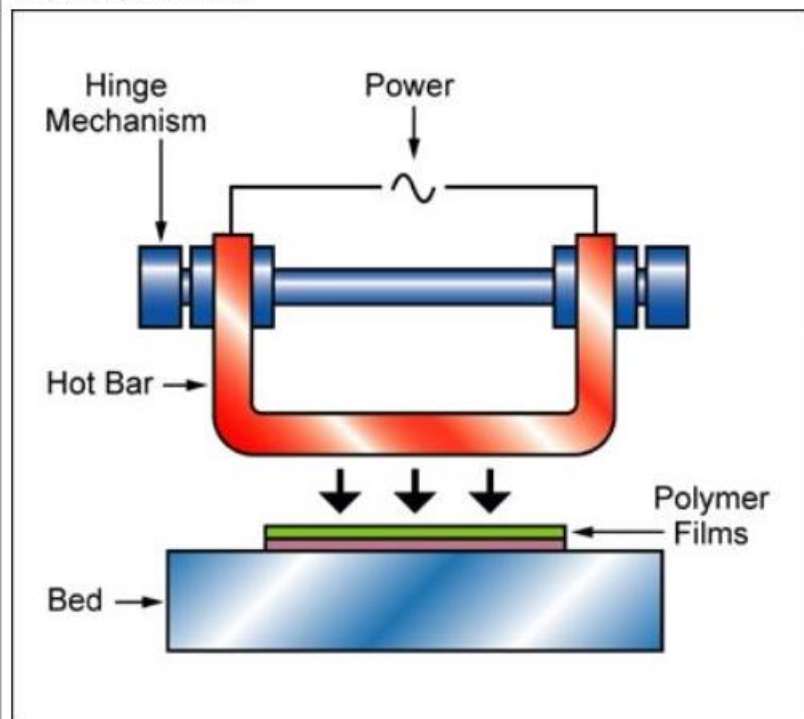
Process schematic



Thermal welding –
polymers – Hot bar
and impulse welding

Hinge mechanism,
power, hot bar, bed,
polymer films

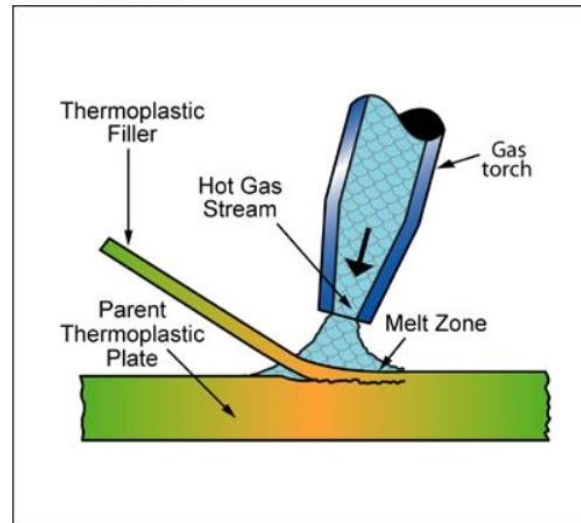
Process schematic



Thermal welding –
polymers – Hot gas
welding

Thermoplastic filler,
hot gas stream, gas
torch, melt zone,
parent thermoplastic
plate

Process schematic

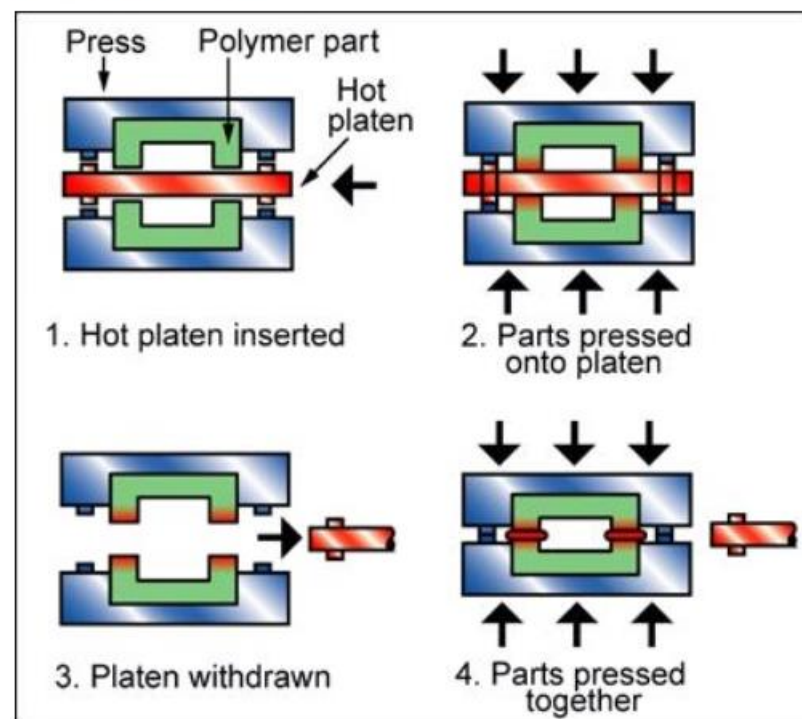


Thermal welding –
polymers – Hot plate
welding

Hot platen inserted,
parts pressed onto
platen, platen
withdrawn, parts
pressed together

Press, polymer part,
hot platen,

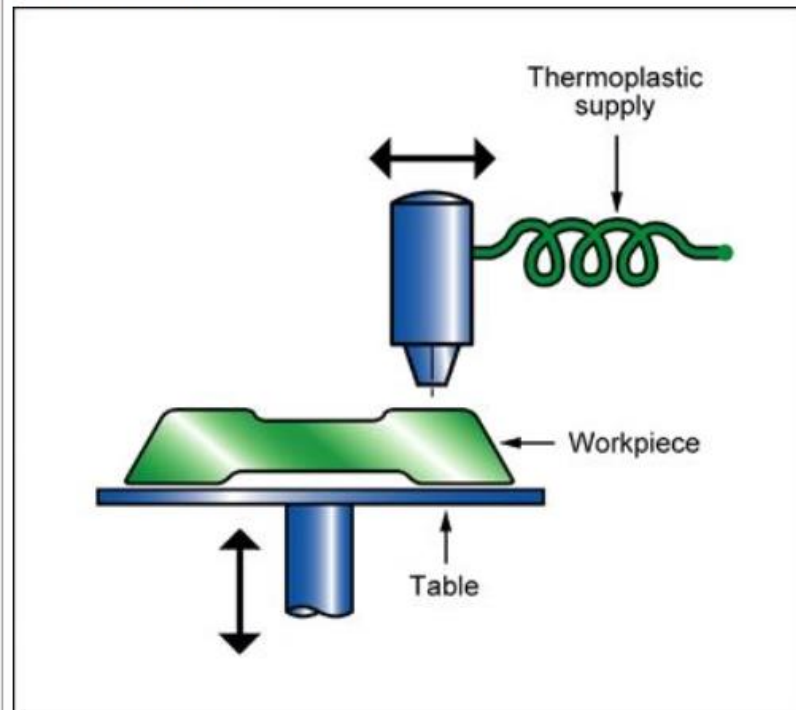
Process schematic



Additive manufacturing – deposition methods – ballistic particle manufacturing

Workpiece, table, thermoplastic supply

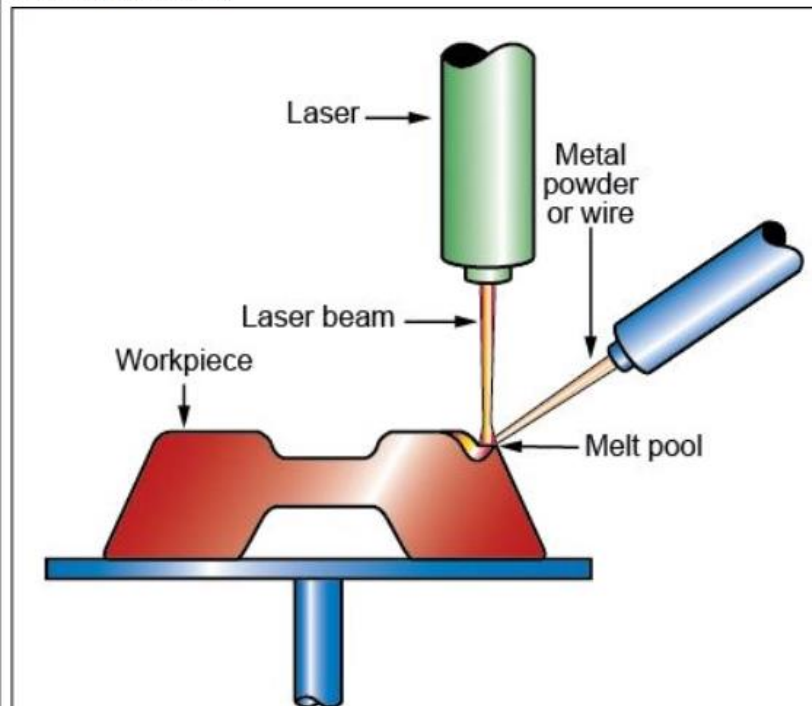
Process schematic



Laser powder forming

Laser, laser beam, workpiece, melt pool, metal powder or wire

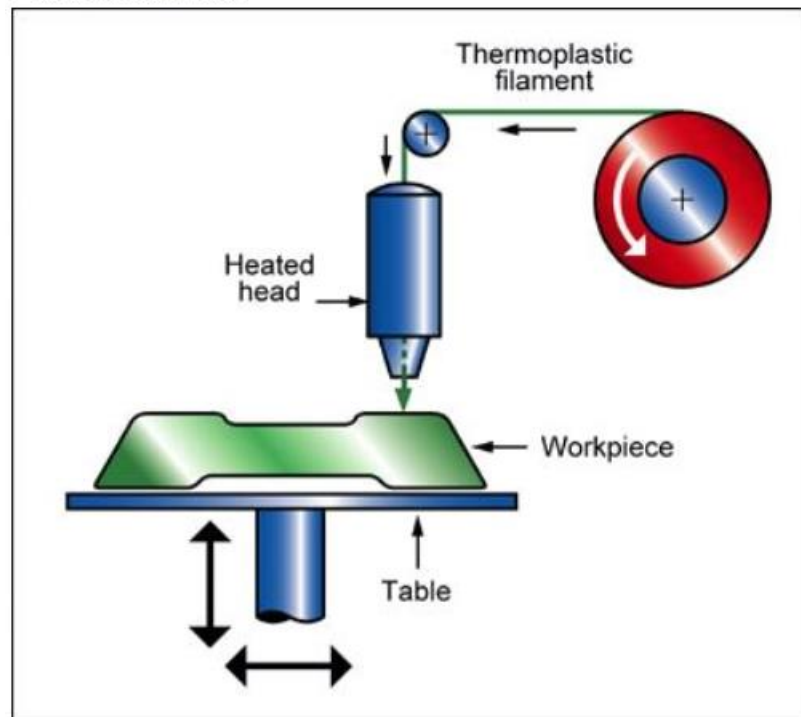
Process schematic



Material extrusion
Fused deposition
modelling

Heated head,
thermoplastic
filament, workpiece,
table

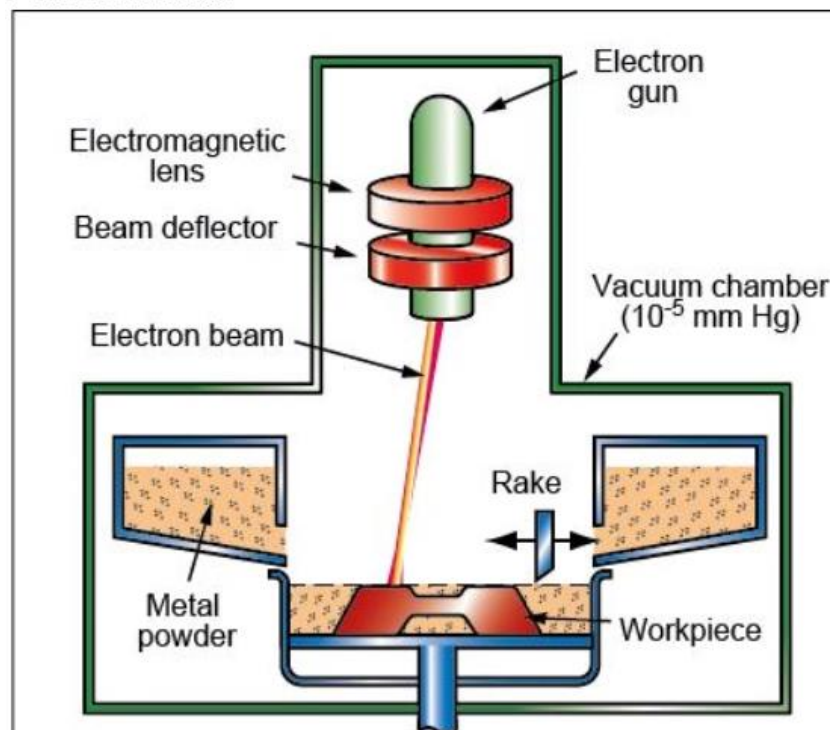
Process schematic



Powder bed fusion –
electron beam melting

Electromagnetic lens,
beam deflector,
electron gun, vacuum
chamber, electron
beam, rake, metal
powder, workpiece

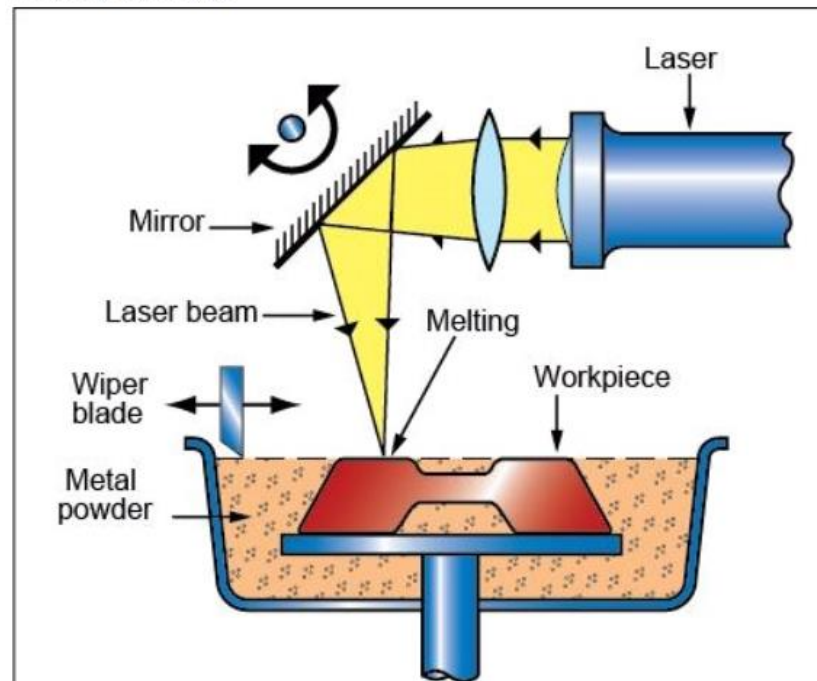
Process schematic



Powder bed fusion – selective laser melting of metals

Mirror, laser, laser beam, melting, wiper blade, metal powder, workpiece

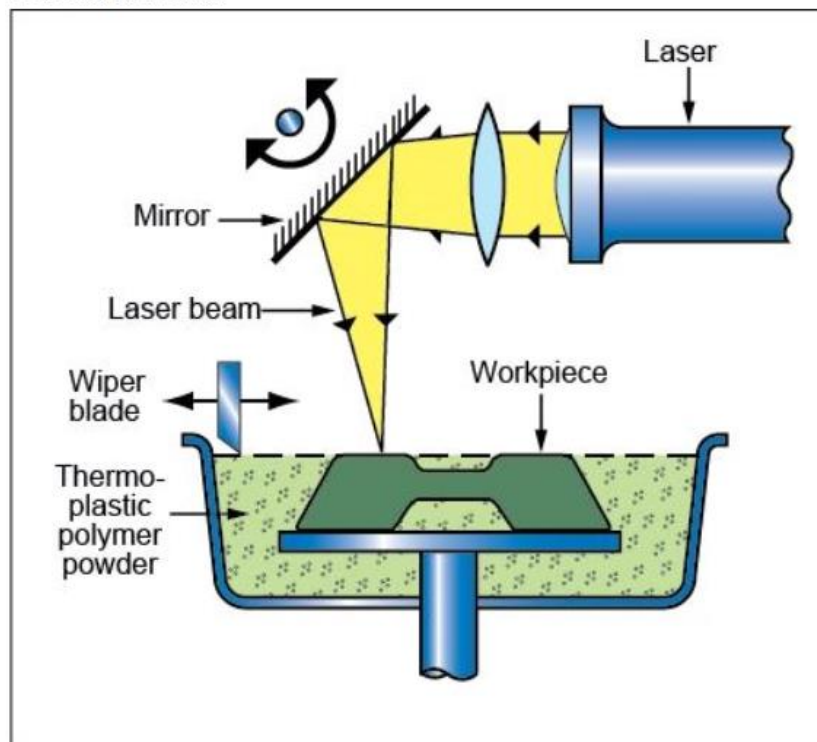
Process schematic



Selective laser sintering of polymers

Mirror, laser, laser beam, wiper blade, workpiece, thermos-plastic polymer powder

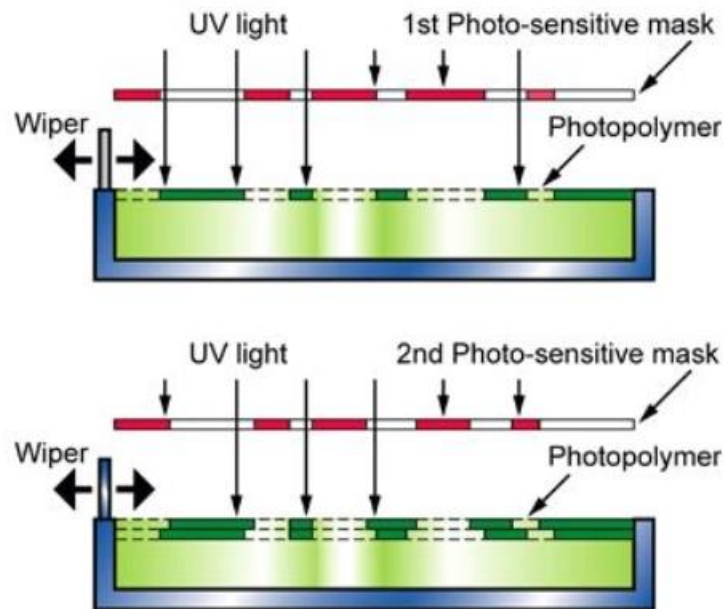
Process schematic



Solid ground curing

Wiper, UV light, 1st
photo-sensitive mask,
photopolymer, 2nd
photo-sensitive mask

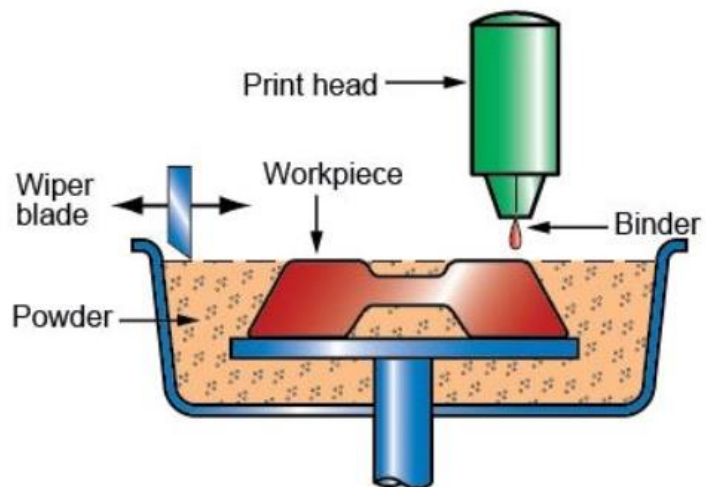
Process schematic



Printing methods –
binder jetting

Print head, workpiece,
wiper blade, powder,
binder

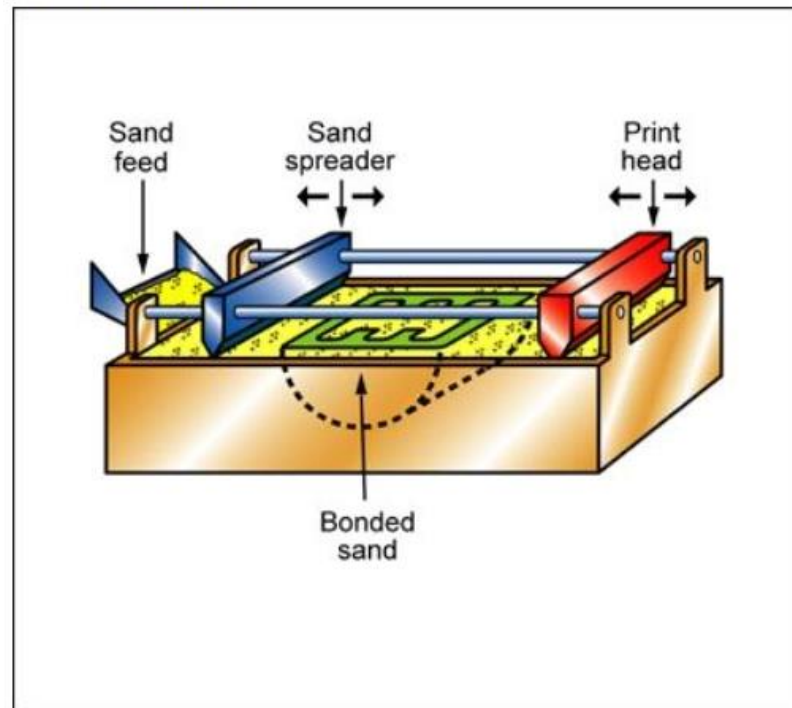
Process schematic



Printing methods –
ceramic moulding
prototyping for metal
casting

Bonded sand, sand
feed, sand spreader,
print head

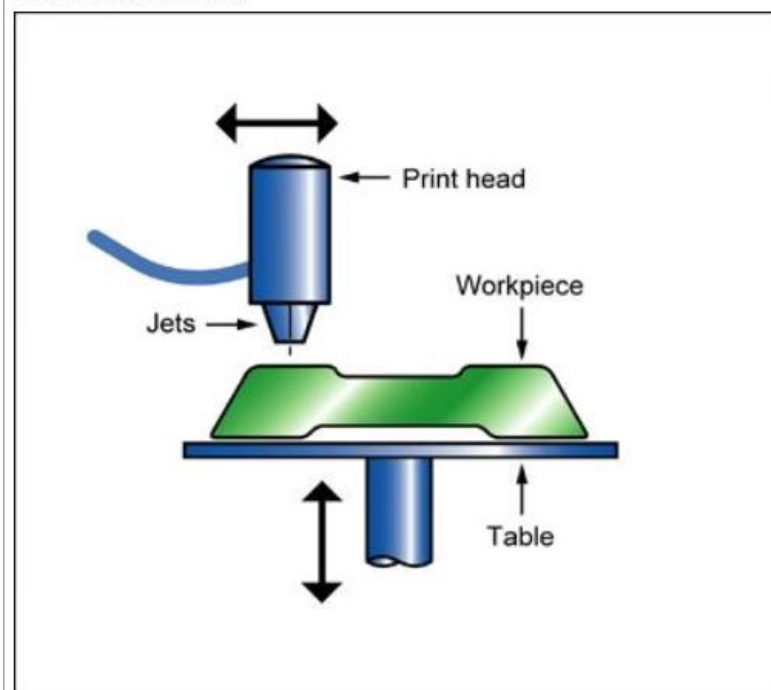
Process schematic



Material jetting
3D printing(?)

Print head, jets,
workpiece, table

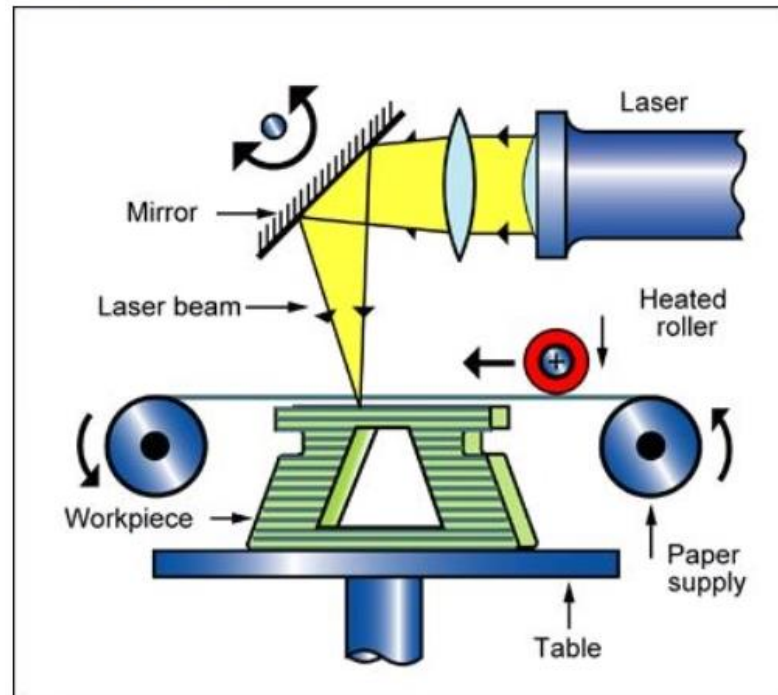
Process schematic



Sheet lamination –
laminated object
manufacture

Mirror, laser, laser
beam, heated roller,
paper supply, table,
workpiece

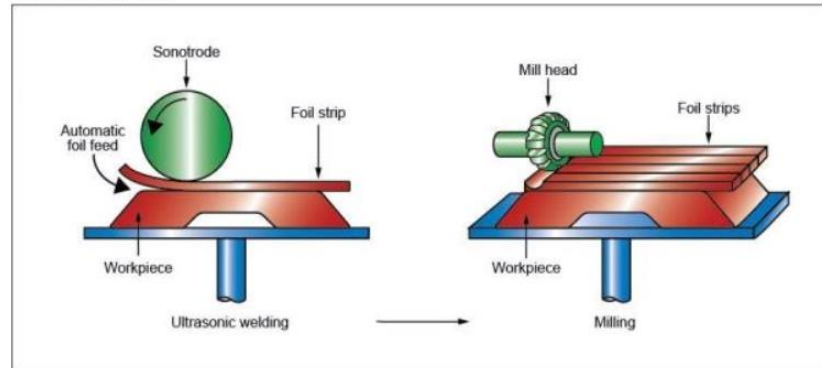
Process schematic



Sheet lamination –
ultrasonic
consolidation

Automatic foil feed,
sonotrode, foil strip,
workpiece, ultrasonic
welding, milling, mill
head, foil strips

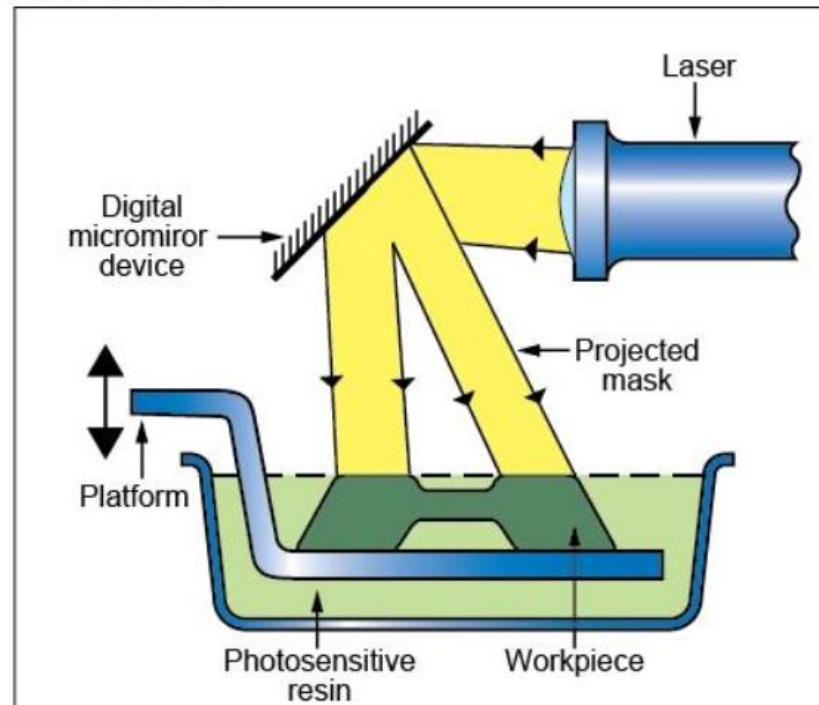
Process schematic



Mask projection
stereolithography

Digital micromirror
device, laser,
platform, projected
mask, workpiece,
photosensitive resin

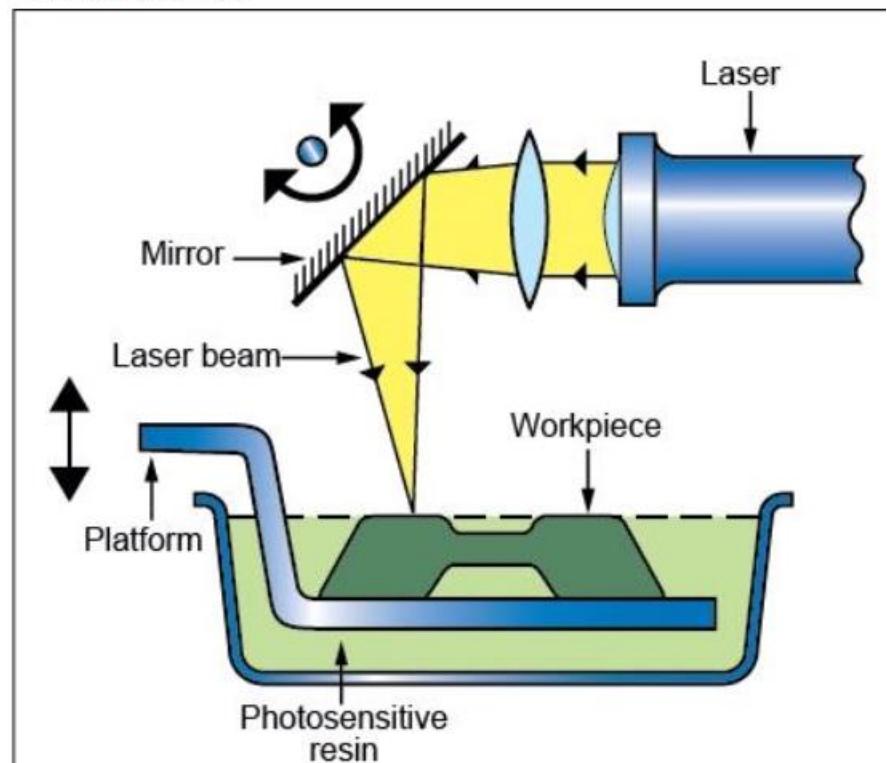
Process schematic



Stereolithography

Mirror, laser, laser
beam, workpiece,
platform,
photosensitive resin

Process schematic

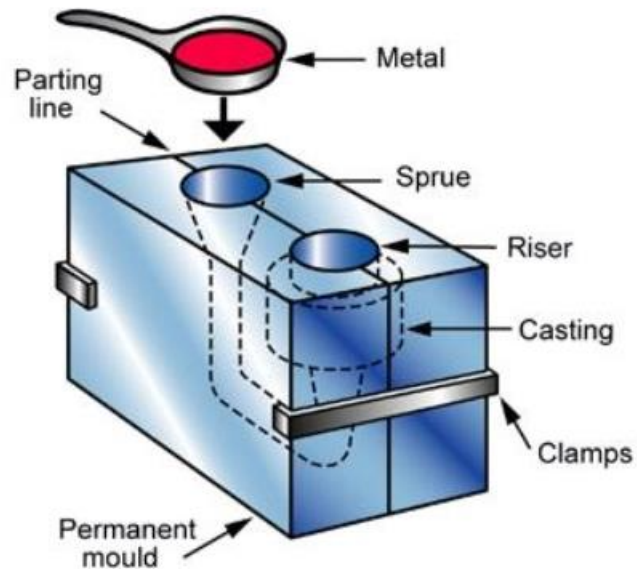


Casting – gravity die casting

Parting line, metal, sprue, riser, casting, clamps, permanent mold

Used to make metallic teapots/kettles

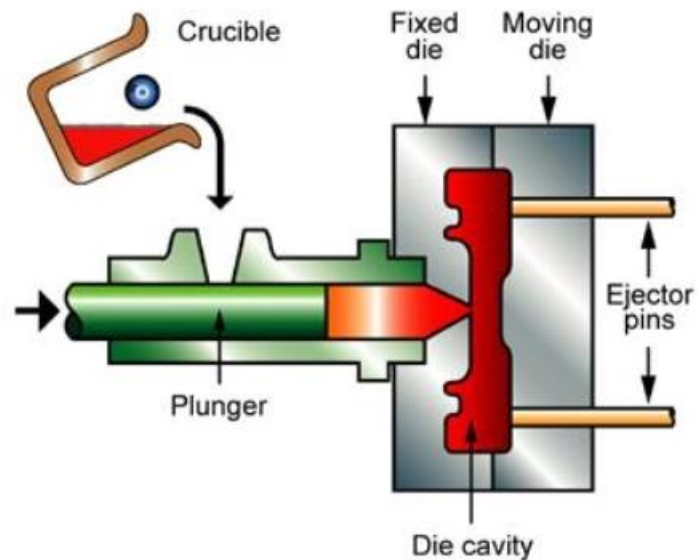
Process schematic



Casting – high pressure die casting

Crucible, plunger, fixed die, moving die, ejector pins, die cavity

Process schematic



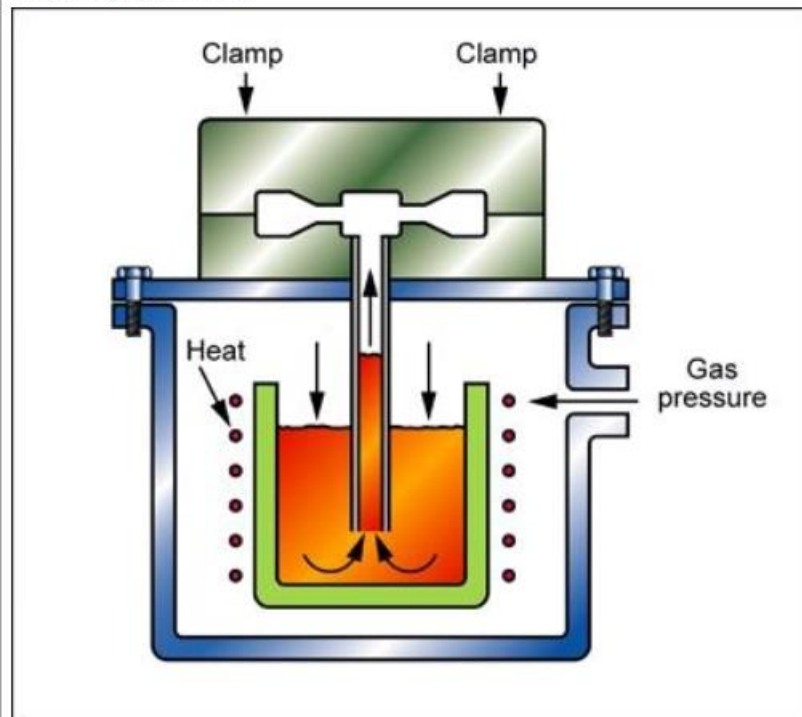
Low pressure die casting

Clamp, heat, gas pressure

Used to make funny hubcap looking silver thingy



Process schematic

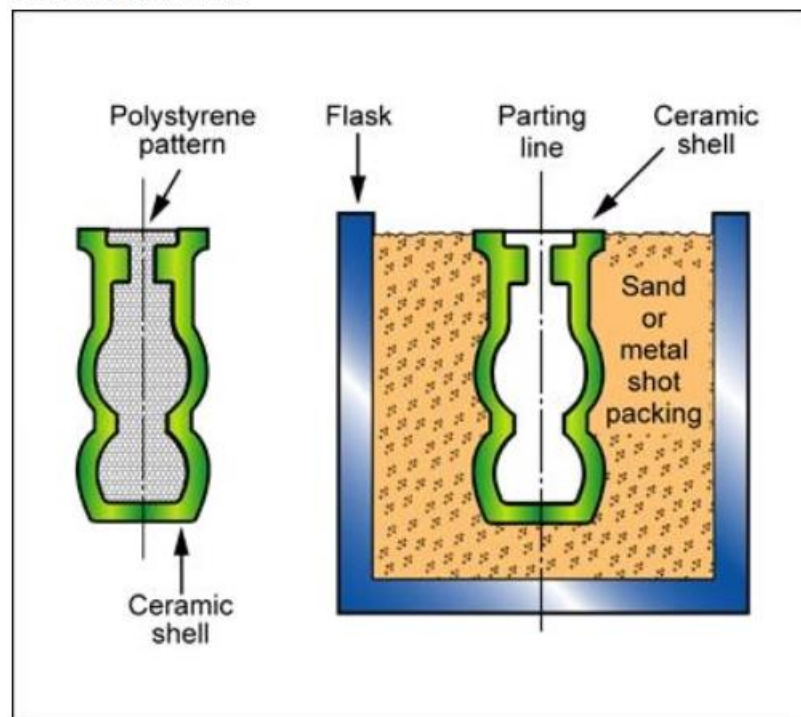


Investment casting – ceramic shell evaporative mould casting

Polystyrene pattern, ceramic shell, flask, parting line, sand or metal shot packing

Used to make taps

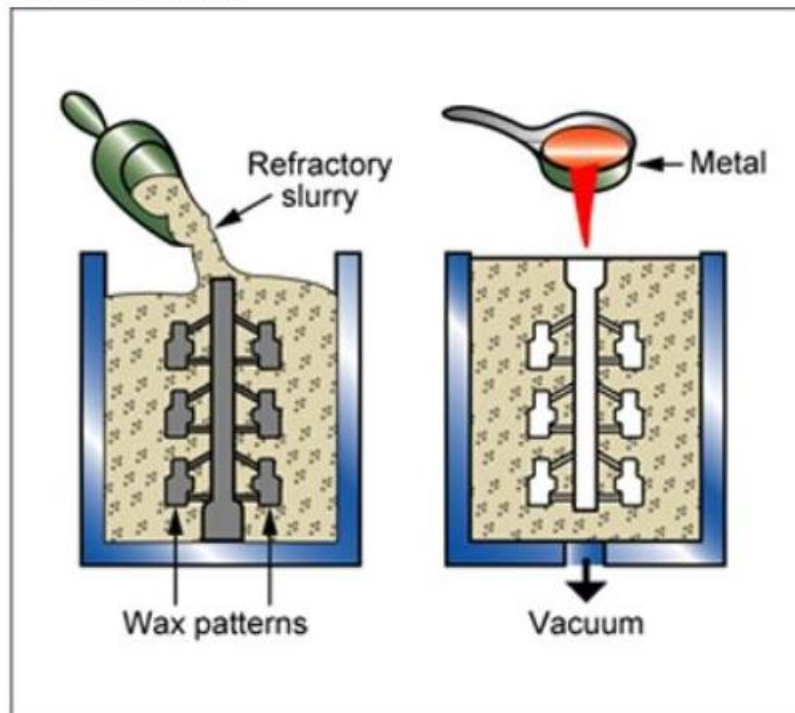
Process schematic



Investment casting

Refractory slurry, wax patterns, metal, vacuum

Process schematic

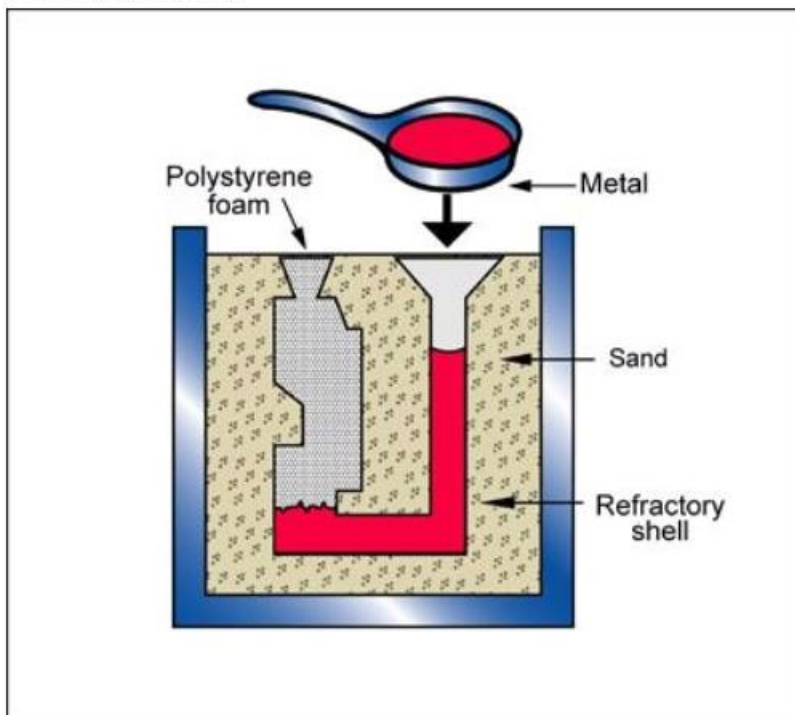


Sand casting – evaporative pattern sand casting

Polystyrene foam,
metal, sand,
refractory shell

Used to make crank
shafts

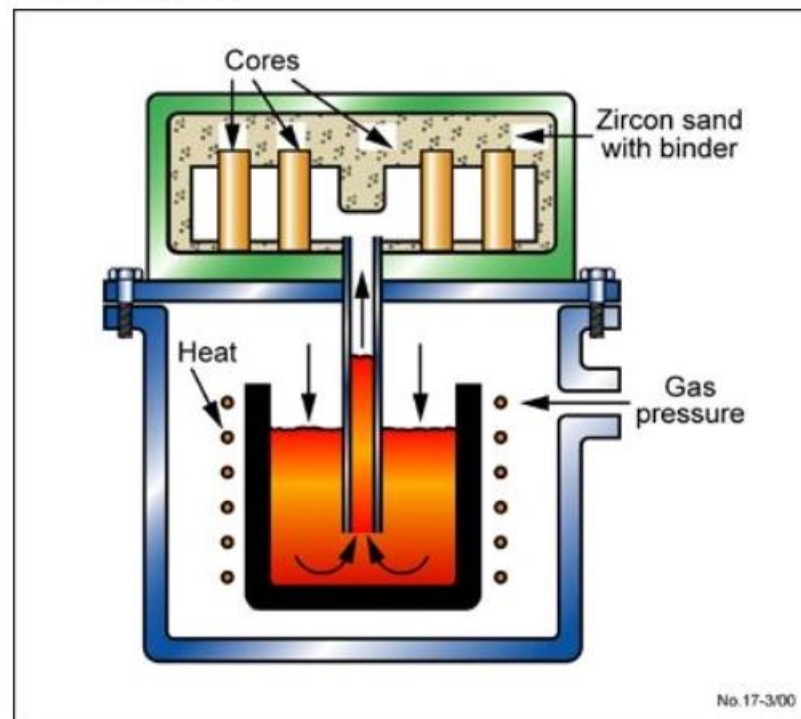
Process schematic



Sand casting – low pressure sand casting

Cores, zircon sand with binder, heat, gas pressure

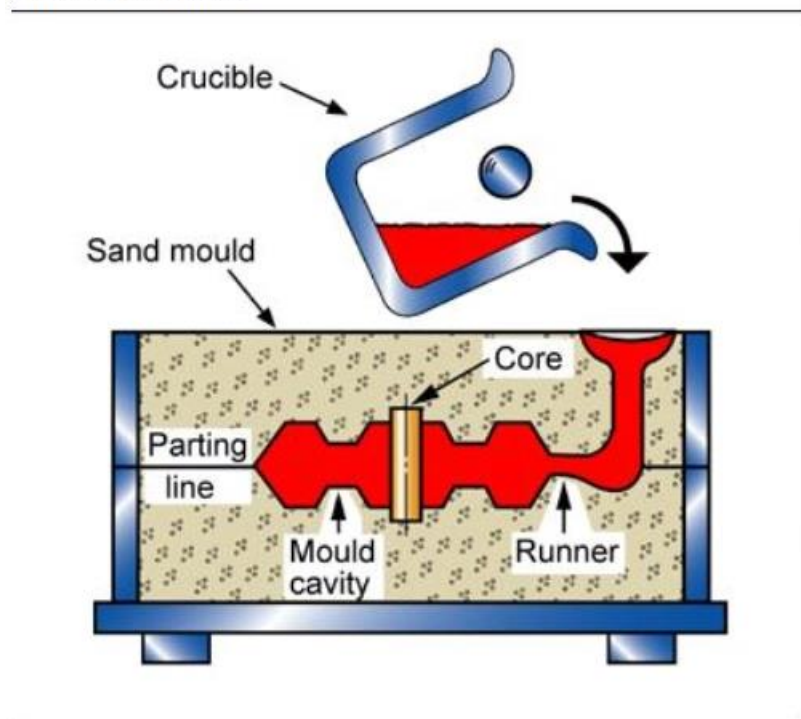
Process schematic



Sand casting

Crucible, sand mould, parting line, core, mould cavity, runner

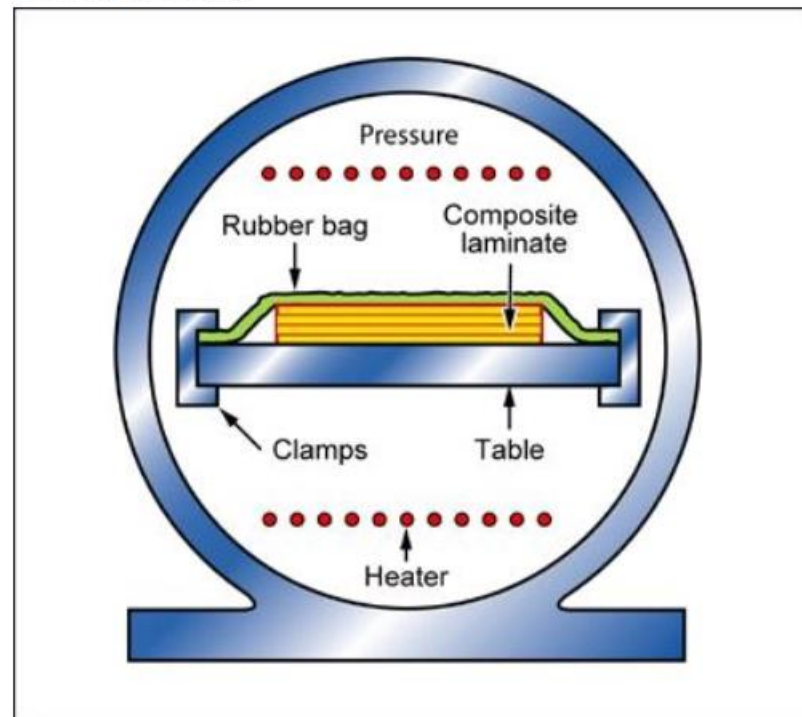
Process schematic



Composite forming – autoclave moulding

Pressure, rubber bag, composite laminate, clamps, table, heater

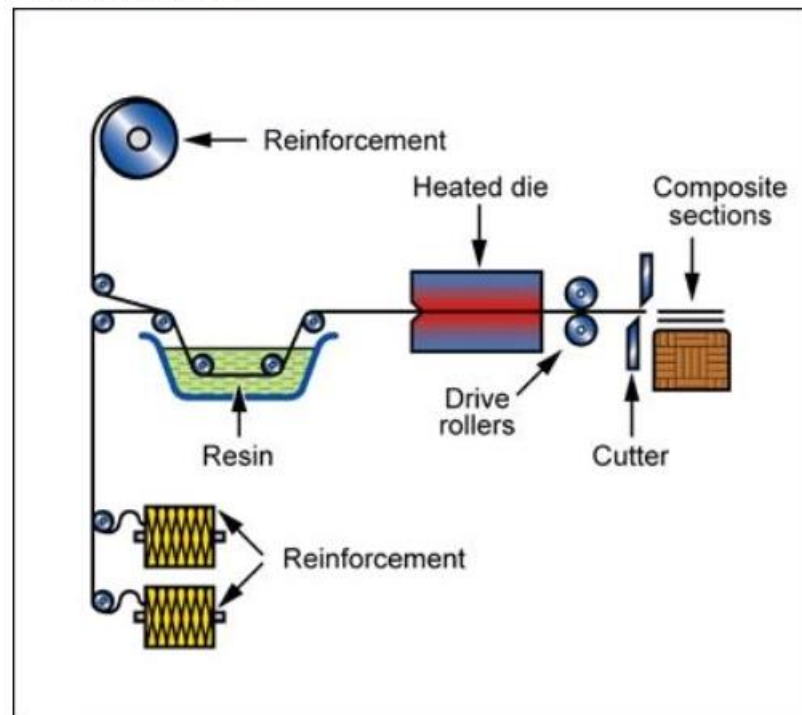
Process schematic



Composite forming – pultrusion

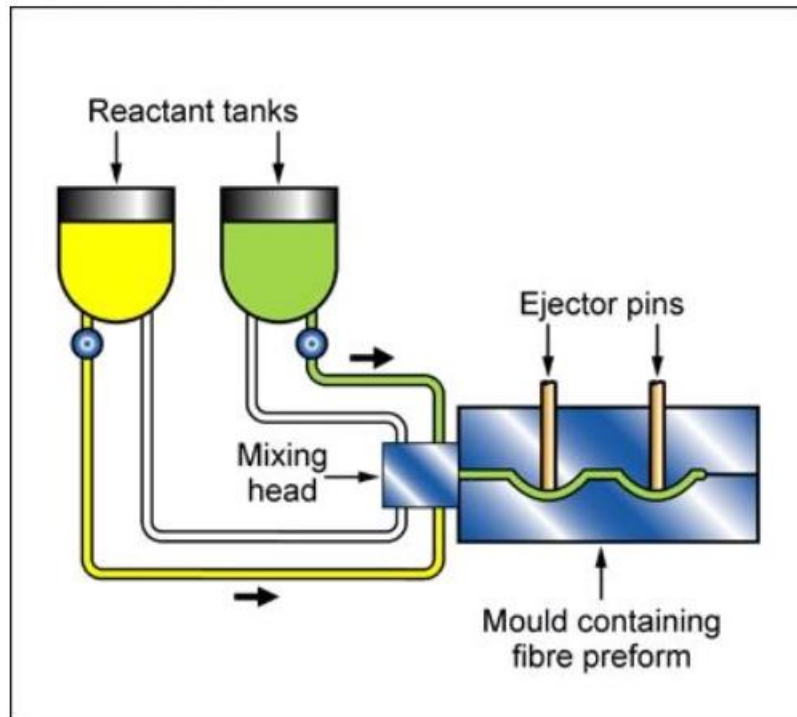
Reinforcement, heated die, composite sections, resin, drive rollers, cutter, reinforcement

Process schematic



Resin transfer moulding (RTM)

Reactant tanks, mixing head, ejector pins, mould containing fibre preform, yellow paint, green paint

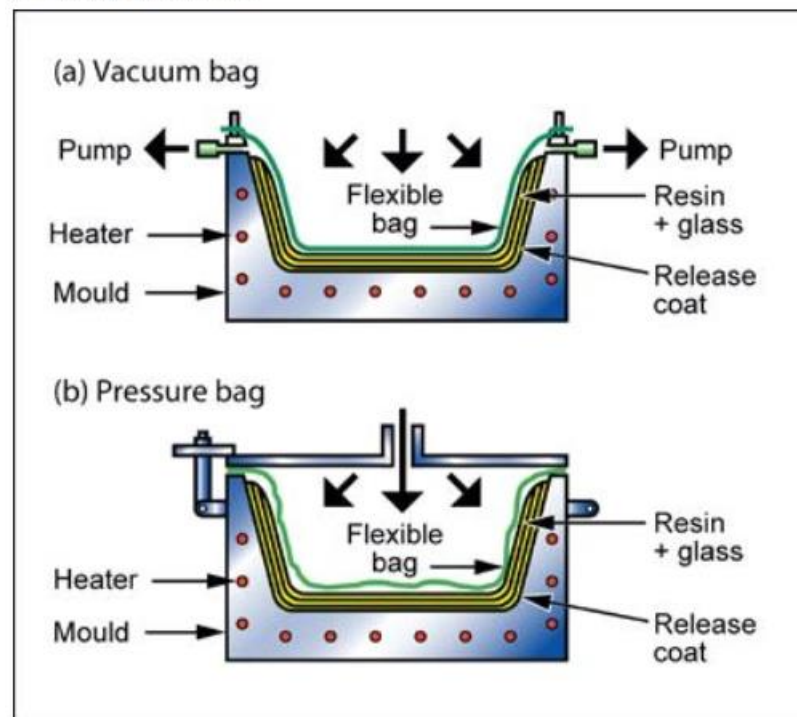


Vacuum pressure bag moulding

Vacuum bag, pressure bag

Pump, heater, mould, flexible bag, resin + glass, release coat,

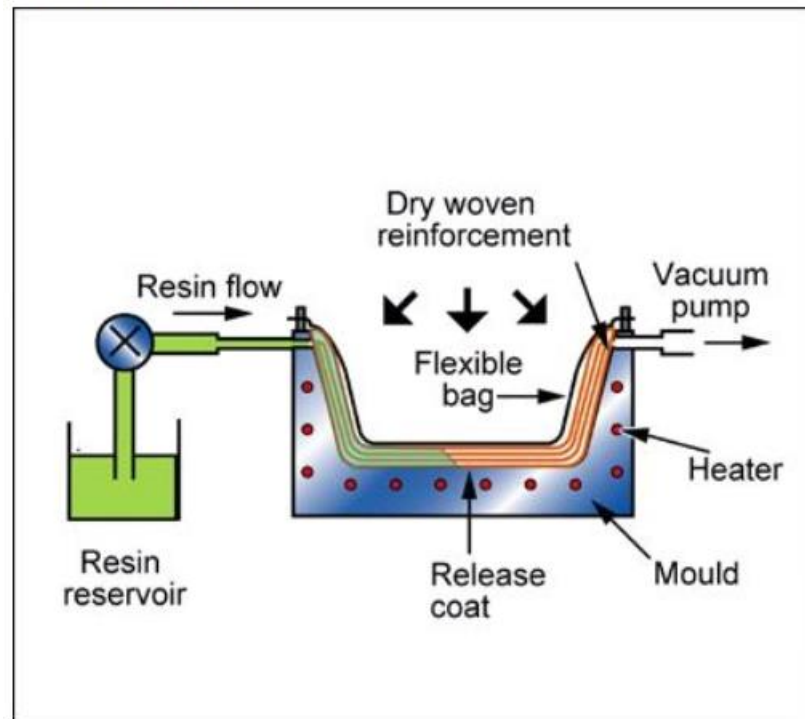
Process schematic



Vacuum assisted
resin transfer
moulding ((rtm)

Resin reservoir, resin
flow, dry woven
reinforcement,
vacuum pump,
flexible bag, release
coat, mould, heater

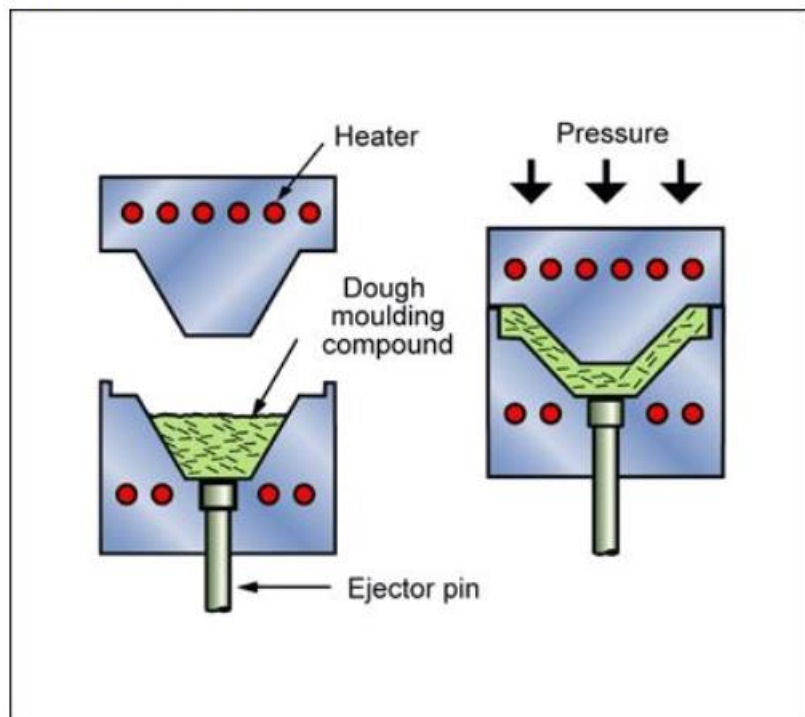
Process schematic



DMC moulding

Heater, dough
moulding compound,
ejector pin, pressure

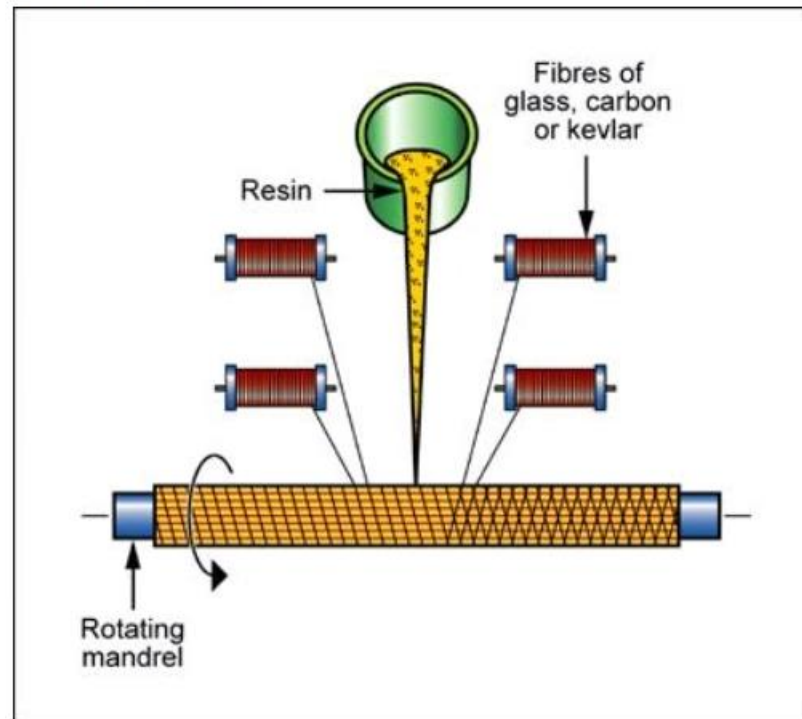
Process schematic



Filament winding

Resin, fibres of glass, carbon or Kevlar, rotating mandrel, sewing, thread, green bucket,

Process schematic



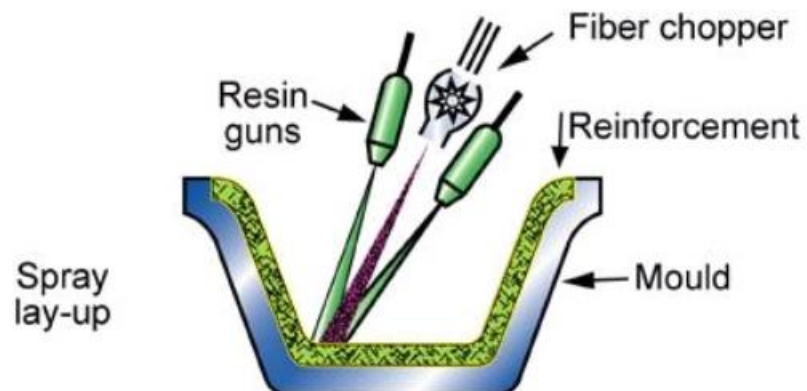
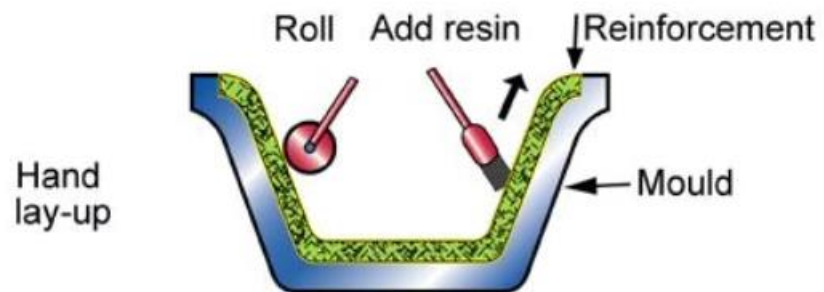
Lay up methods

Hand lay-up

Spray lay-up

Roll, add resin, reinforcement, mould, resin guns, fibre chopper, fiber chopper, reinforcement

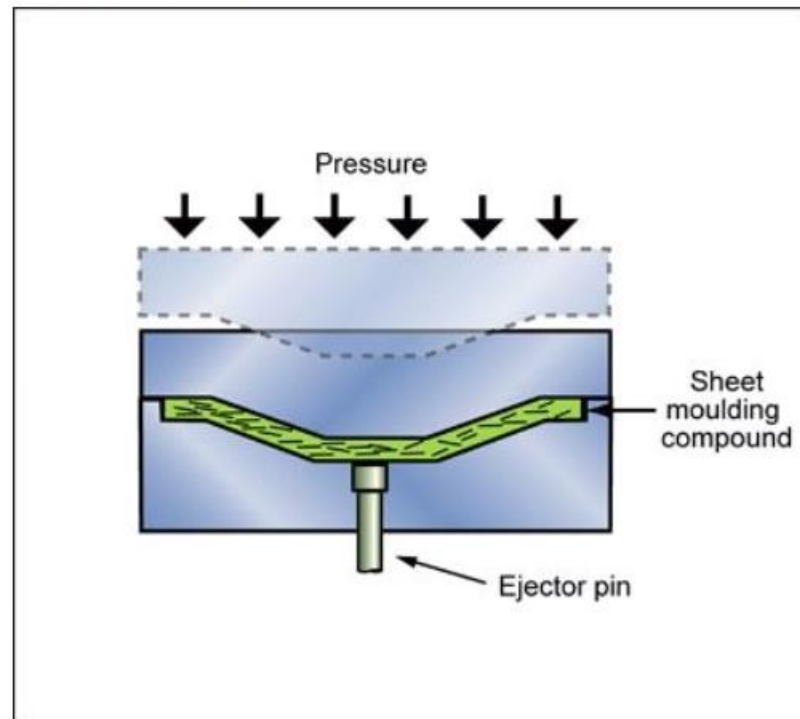
Process schematic



Smc moulding

Pressure, ejector pin,
sheet moulding compound

Process schematic



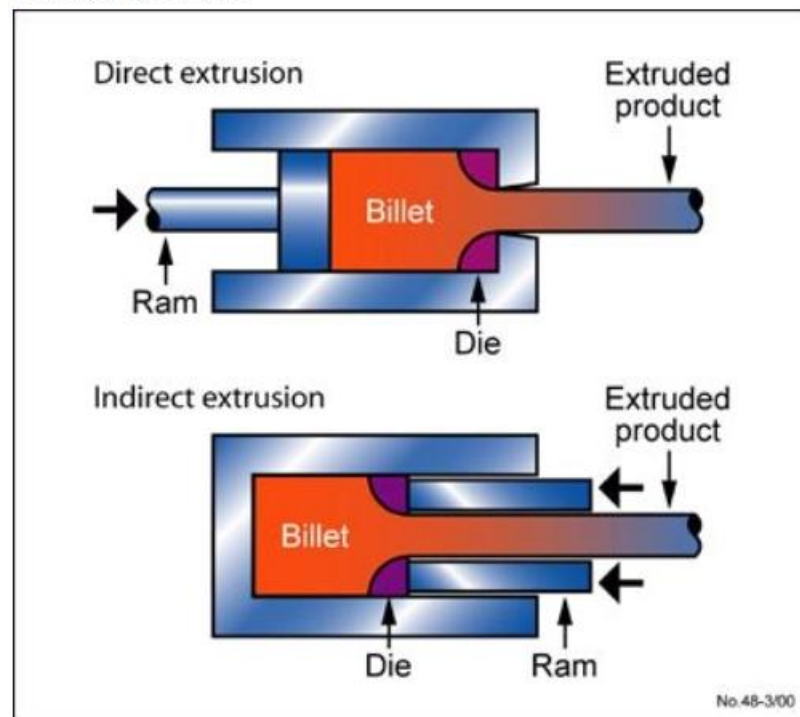
Extrusion

Direct Extrusion
Indirect Extrusion

Ram, billet, die,
extruded product

Used to make rubber
pipes

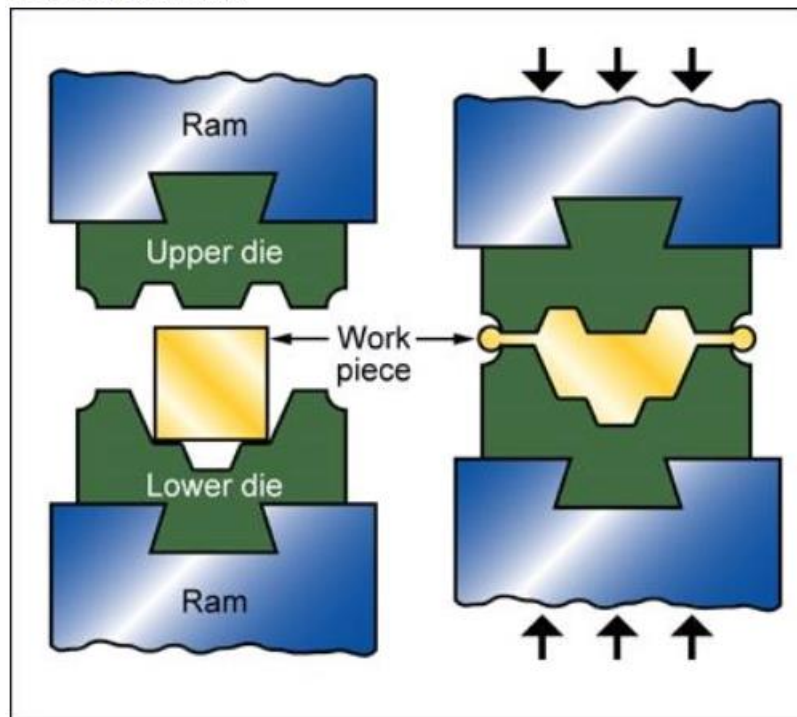
Process schematic



Forging

Ram, upper die, work piece, lower die

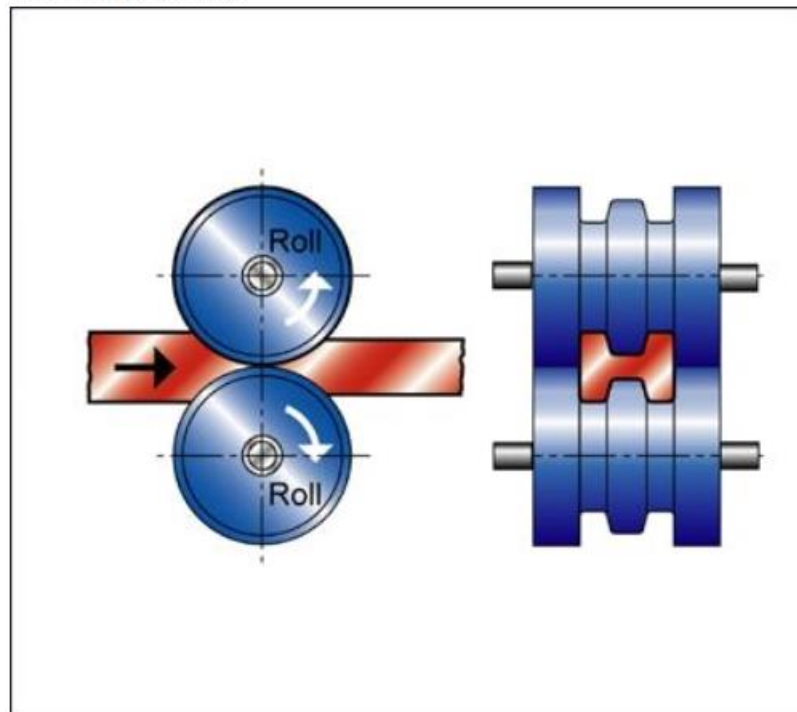
Process schematic



Rolling

Roll

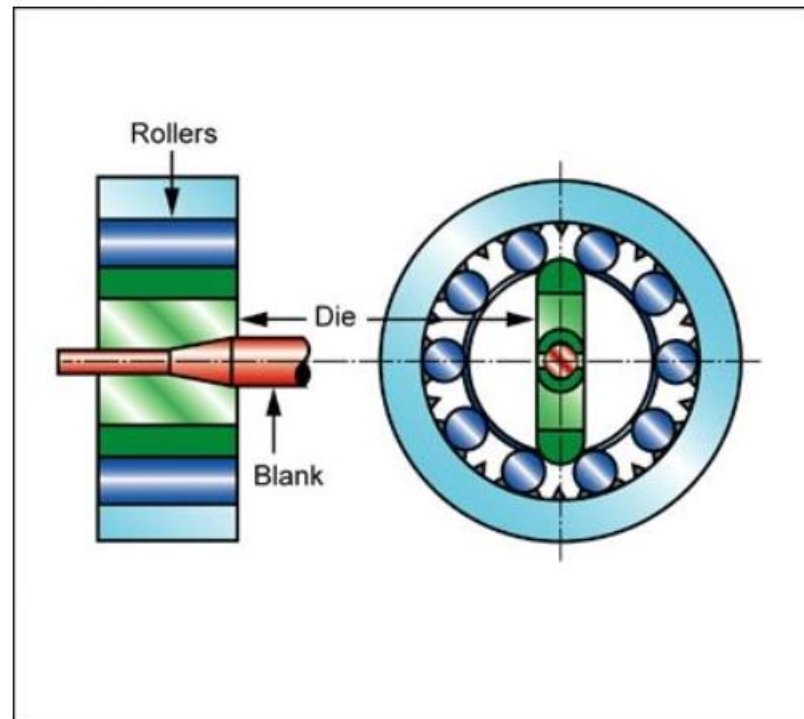
Process schematic



Swaging

Rollers, die, blank

Process schematic

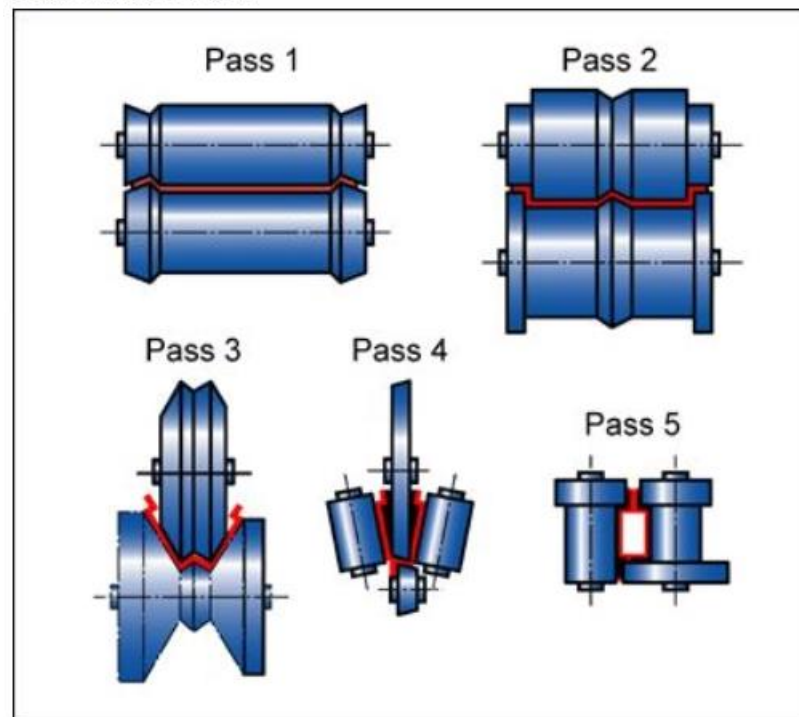


Sheet forming

Pass 1, Pass 2, Pass 3,
Pass 4, Pass 5

Blue rollers, rail, train,
rolling pin

Process schematic



Sheet stamping

Deep drawing

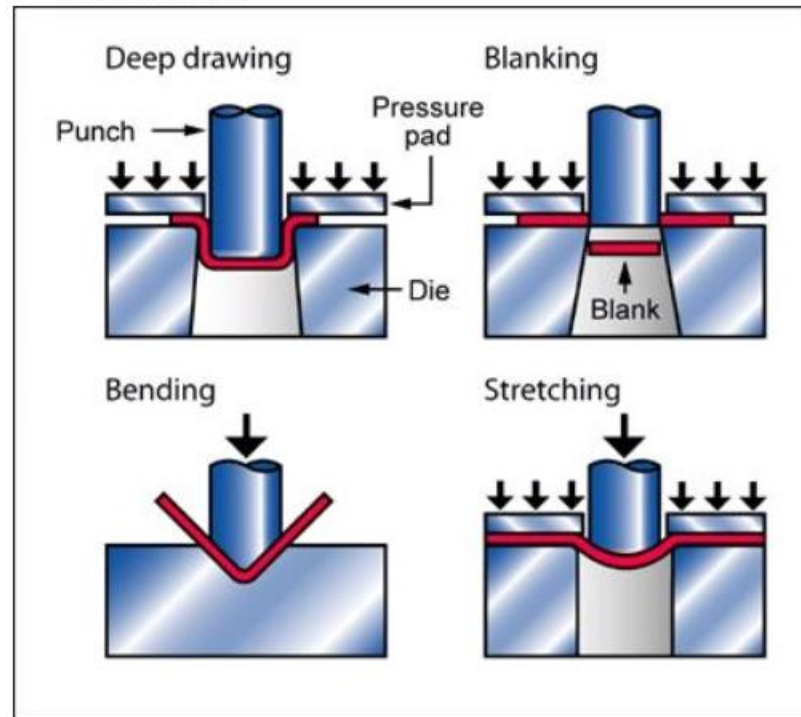
Blanking

Bending

Stretching

Punch, pressure pad,
die, blank

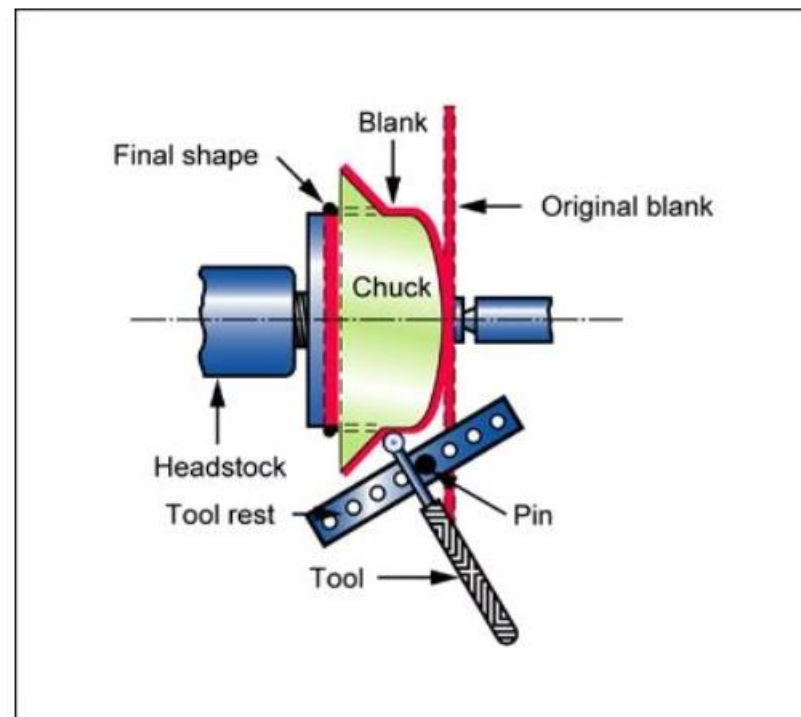
Process schematic



Spinning

Final shape, blank,
original blank, chuck,
headstock, tool rest,
tool, pin

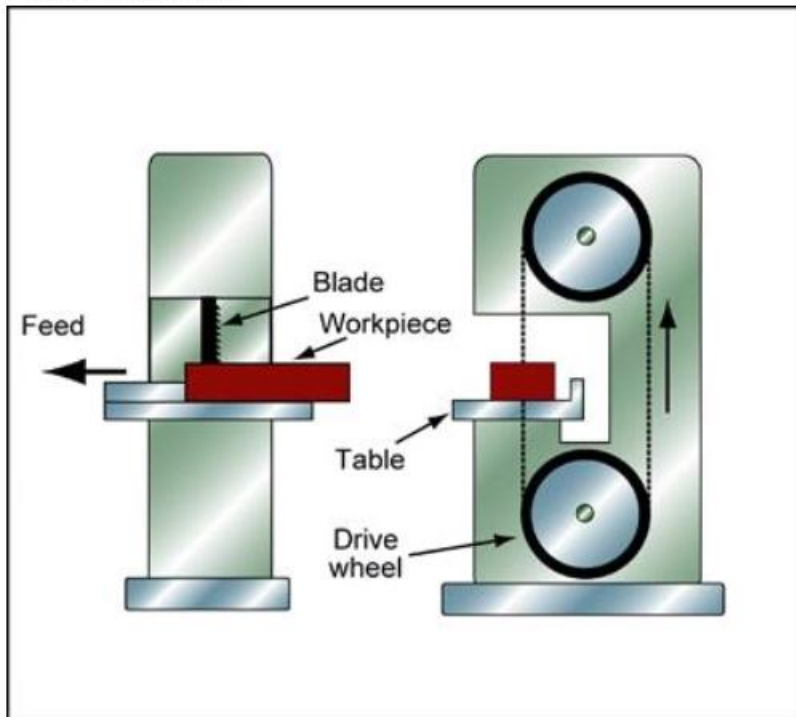
Process schematic



Band sawing

Feed, blade,
workpiece, table, drive
wheel

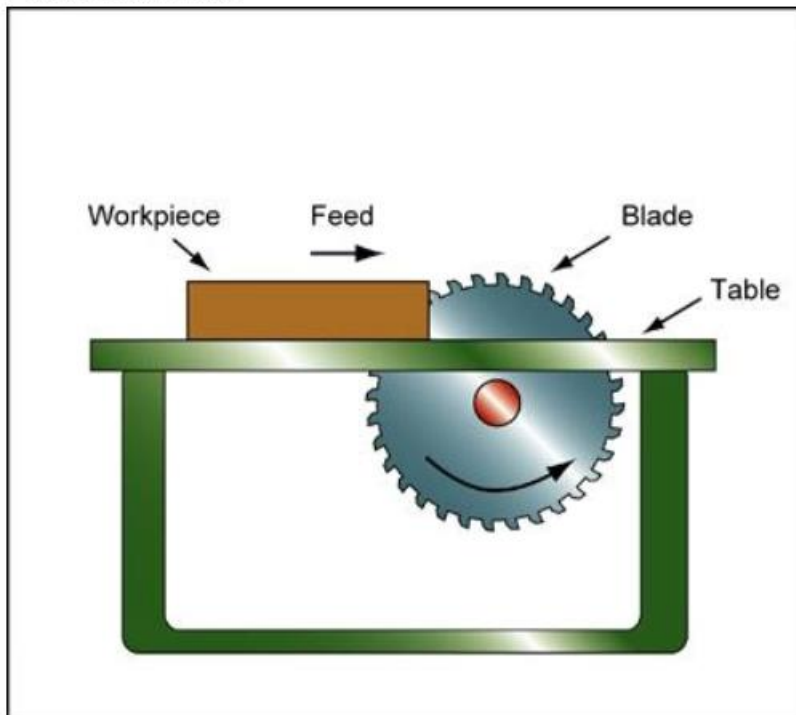
Process schematic



Circular sawing

Workpiece, feed,
blade, table

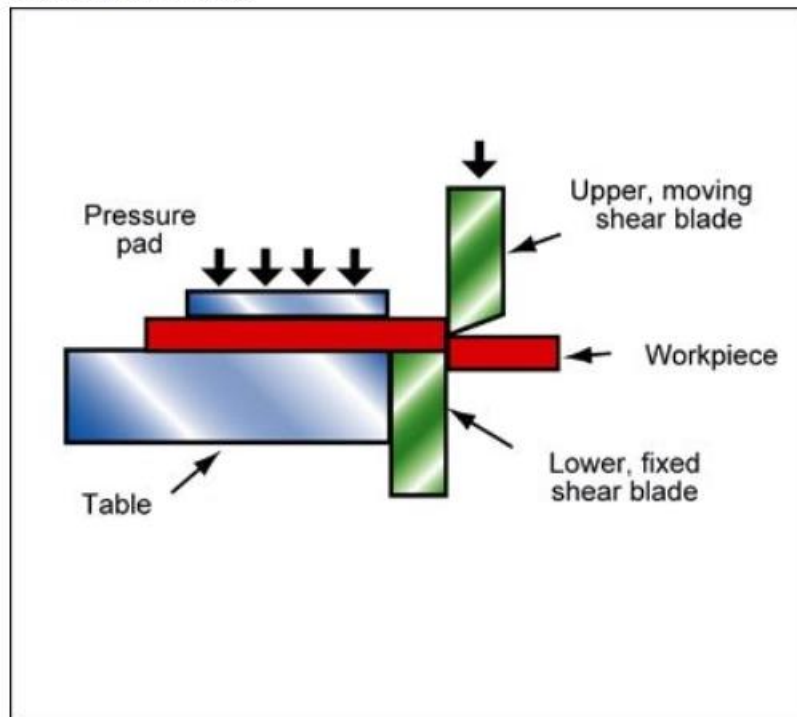
Process schematic



Cropping and
guillotining

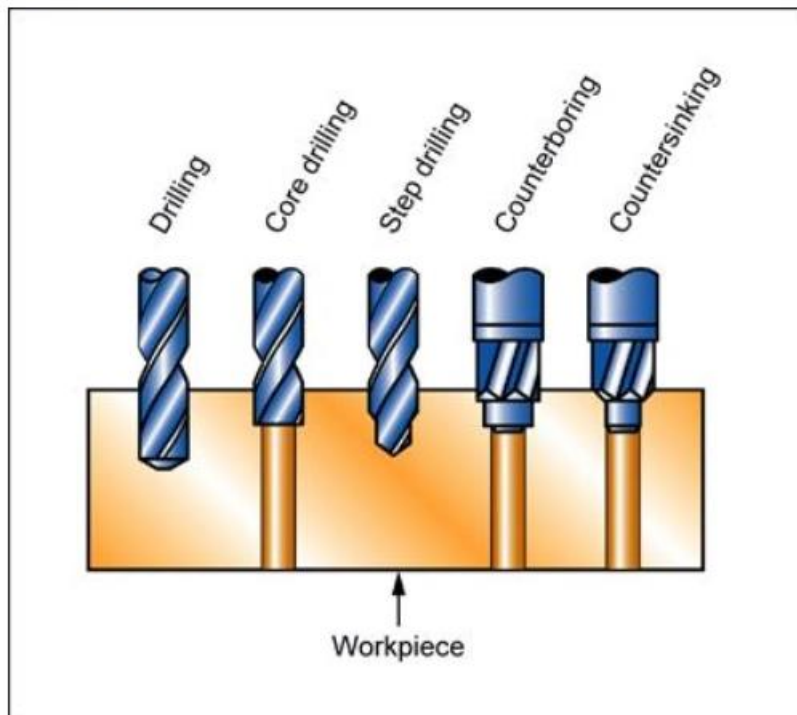
Pressure pad, table,
upper moving shear
blade, lower fixed
shear blade,
workpiece

Process schematic



Drilling
Core drilling
Step drilling
Counterboring
Countersinking

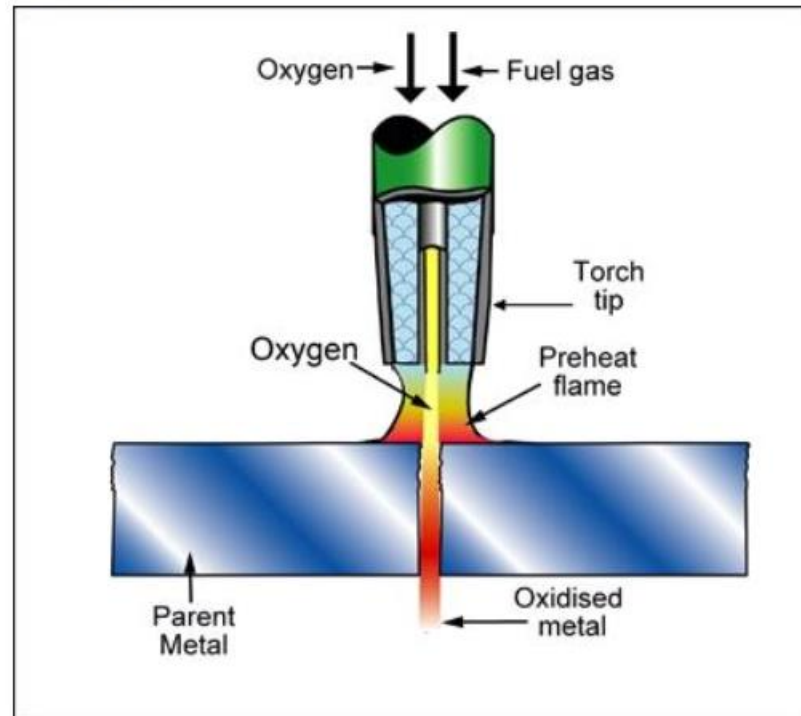
Process schematic



Flame cutting

Oxygen, fuel gas,
torch tip, preheat
flame, parent metal,
oxidised metal

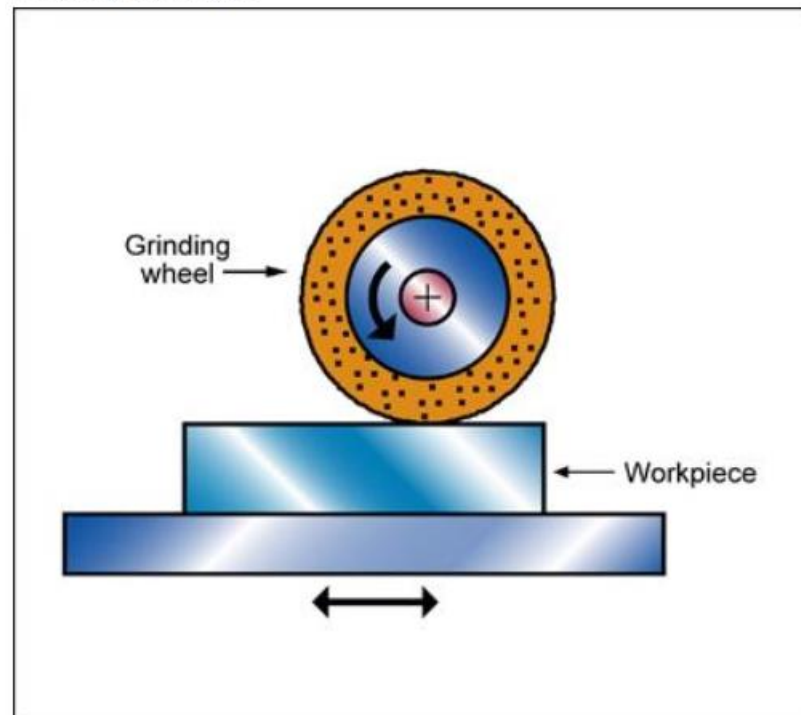
Process schematic



Grinding

Grinding wheel

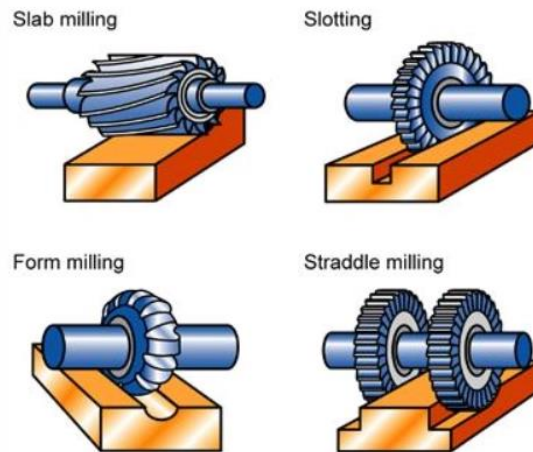
Process schematic



Milling

Slab milling, slotting,
form milling, straddle
milling

Process schematic

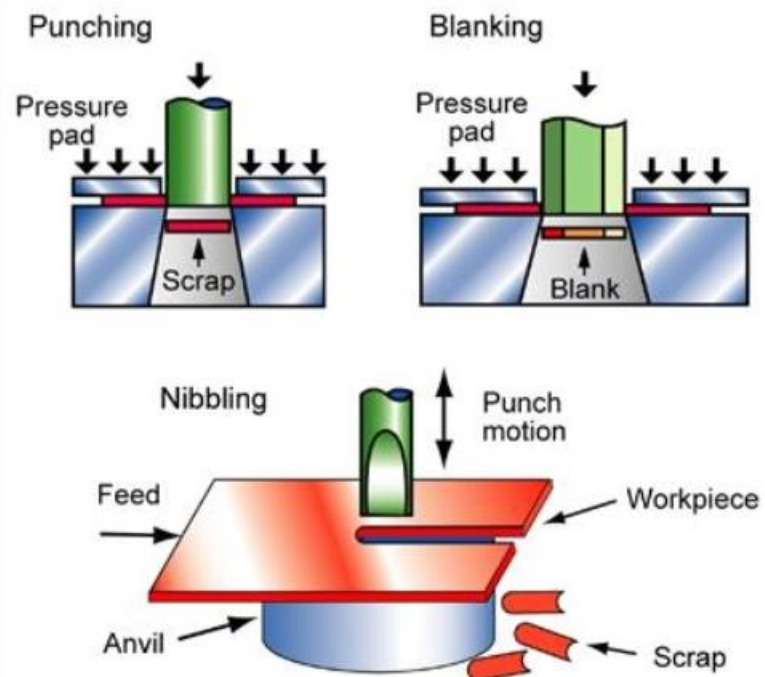


Punching

Blanking, Nibbling, I
LOVE NIBBLING

Punch motion, feed,
anvil, workpiece, scrap,
pressure pad

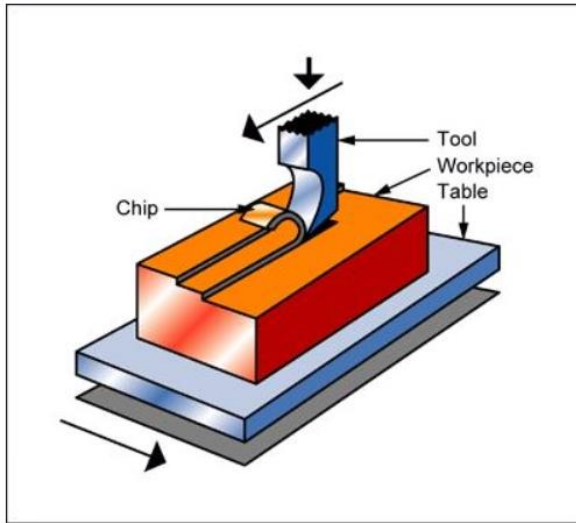
Process schematic



Shaping

Chip, tool, workpiece,
table

Process schematic



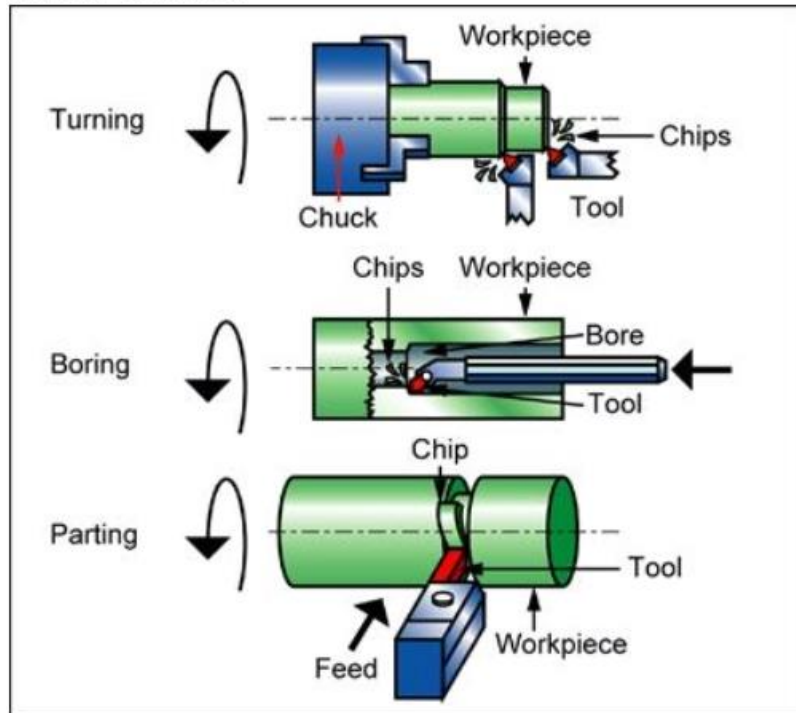
Turning

Boring (like you)

Parting

Workpiece, chuck, tool,
chips, bore, feed, chip

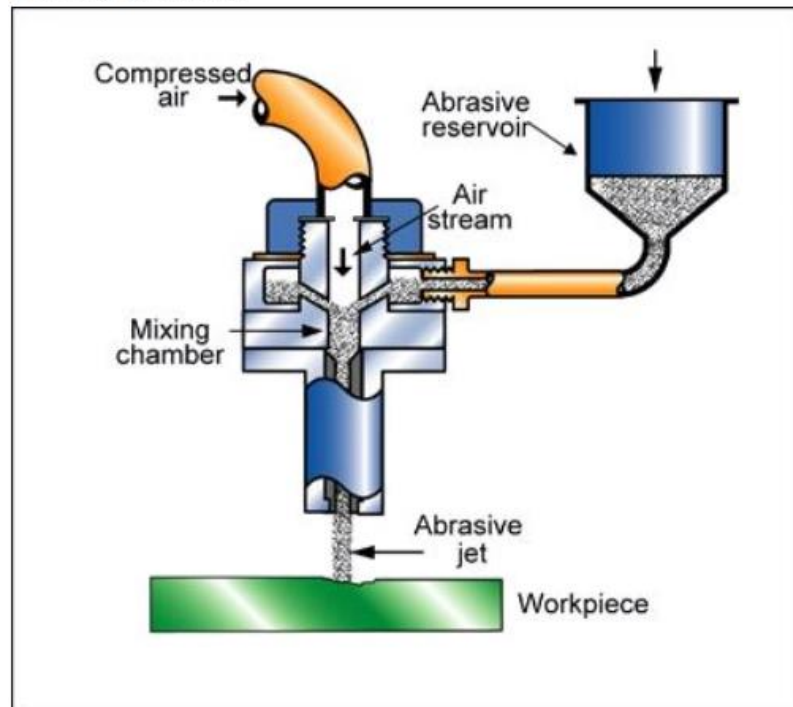
Process schematic



Abrasive jet machining

Compressed air,
abrasive reservoir, air
stream, mixing
chamber, abrasive jet,
workpiece

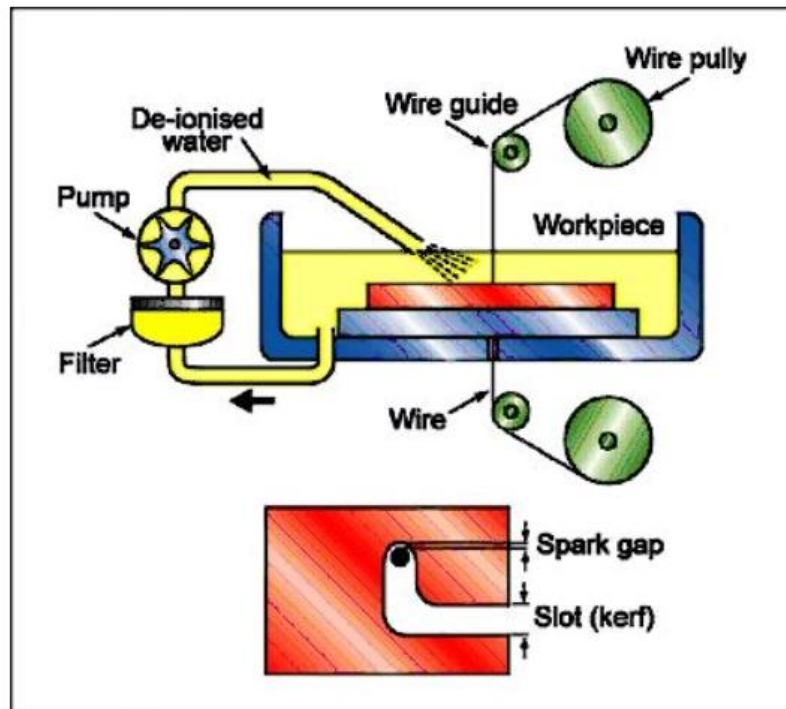
Process schematic



Electric-discharge wire cutting

De-ionised water,
pump, filter, wire guide,
wire pulley, workpiece,
wire, spark gap, slot
(kerf)

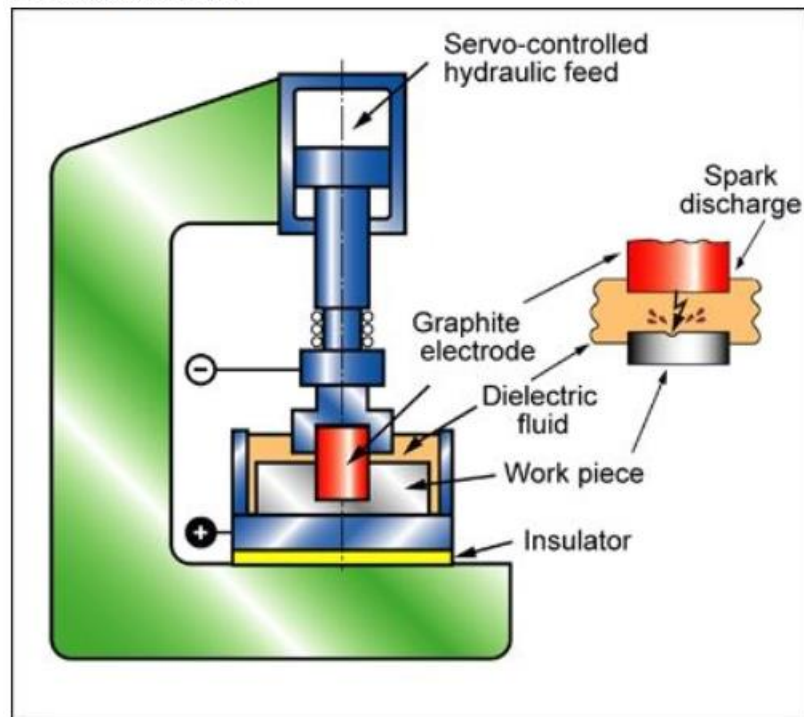
Process schematic



Electric-discharge wire machining

Servo-controlled hydraulic feed, spark discharge, graphite electrode, dielectric fluid, work piece, insulator

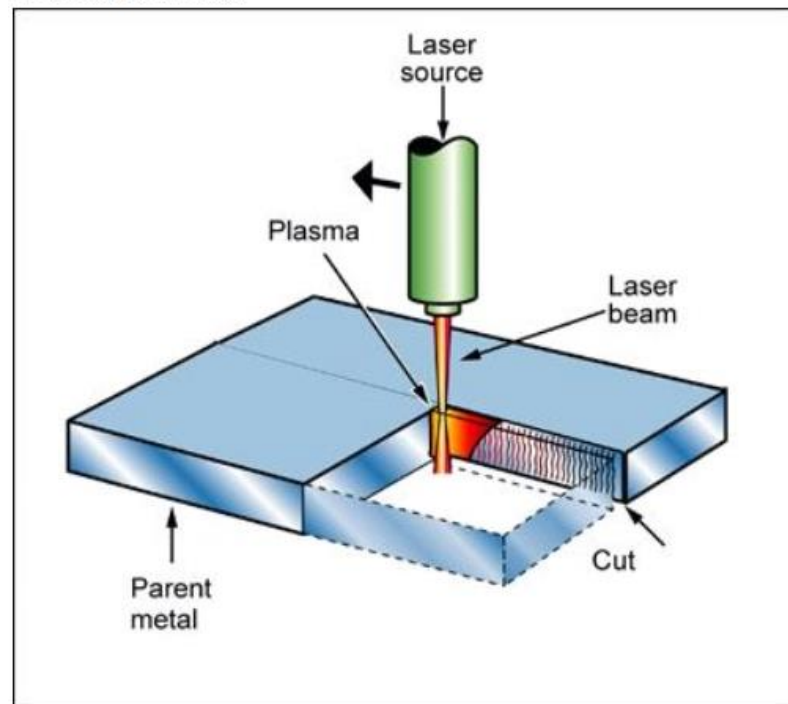
Process schematic



Laser cutting

Laser source, plasma, laser beam, parent metal, cut

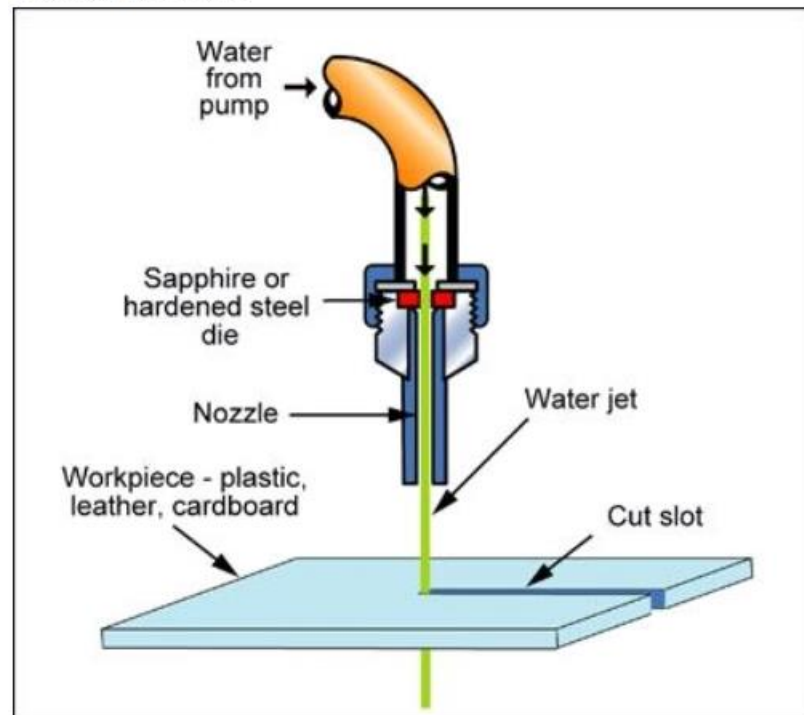
Process schematic



Water-jet cutting

Water from pump,
sapphire or hardened
steel die, nozzle,
workpiece – plastic,
leather, cardboard
Water jet, cut slot

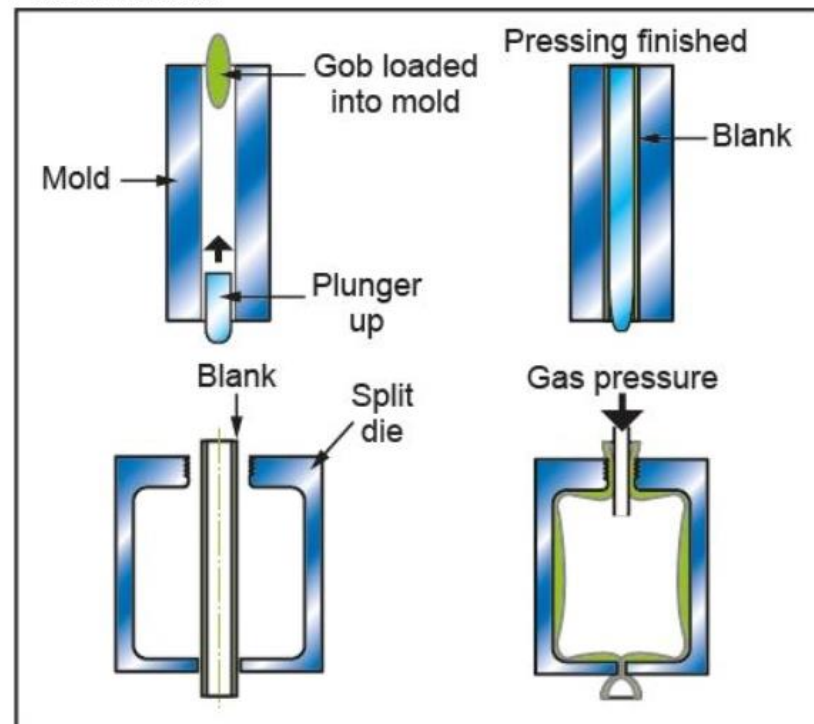
Process schematic



Water-jet cutting

Mold, gob loaded into
mold, plunger up, blank,
split die, gas pressure,
blank, pressing finished

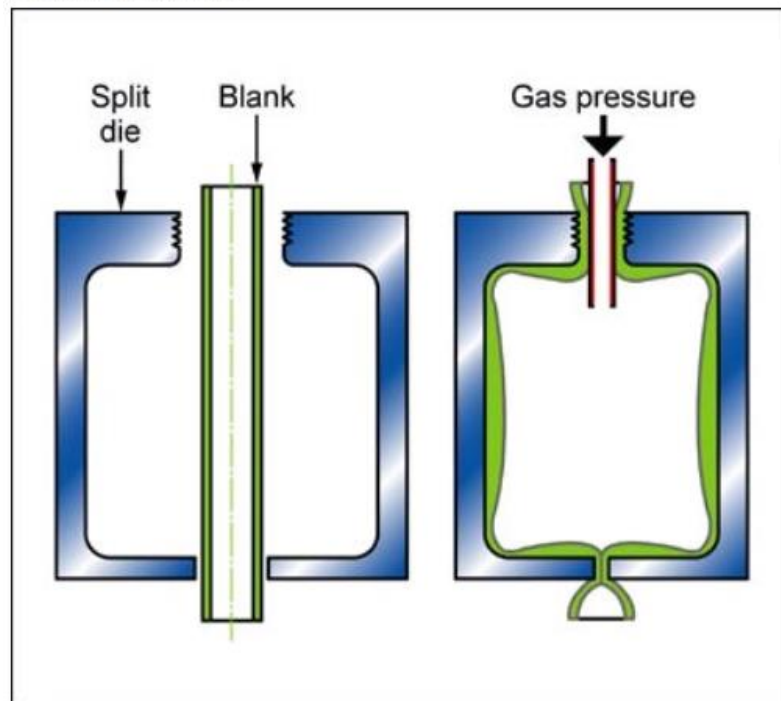
Process schematic



Moulding – ceramic -
blow moulding

Glass, split die, blank,
gas pressure

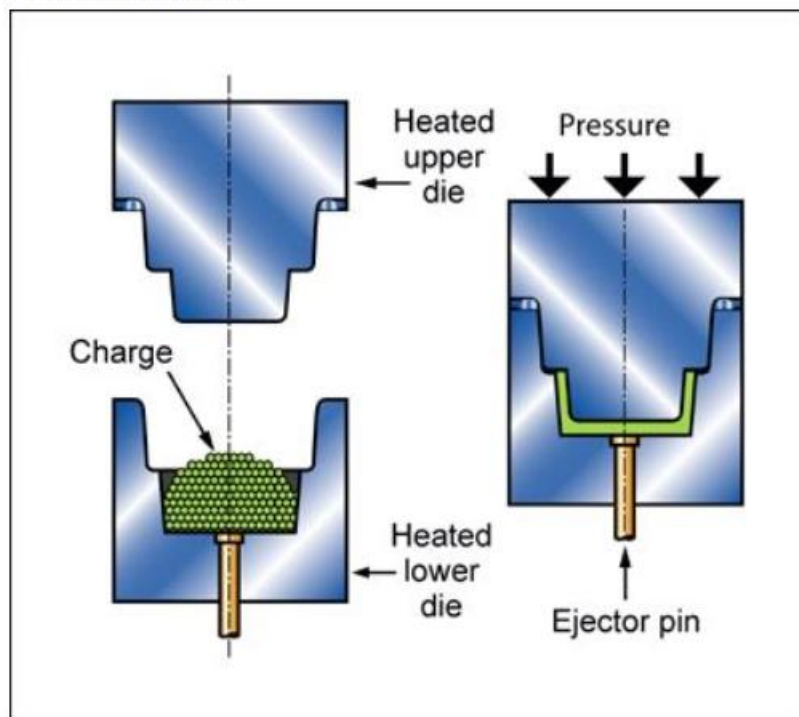
Process schematic



Compression moulding

Heated upper die,
pressure, charge,
heated lower die,
ejector pin

Process schematic

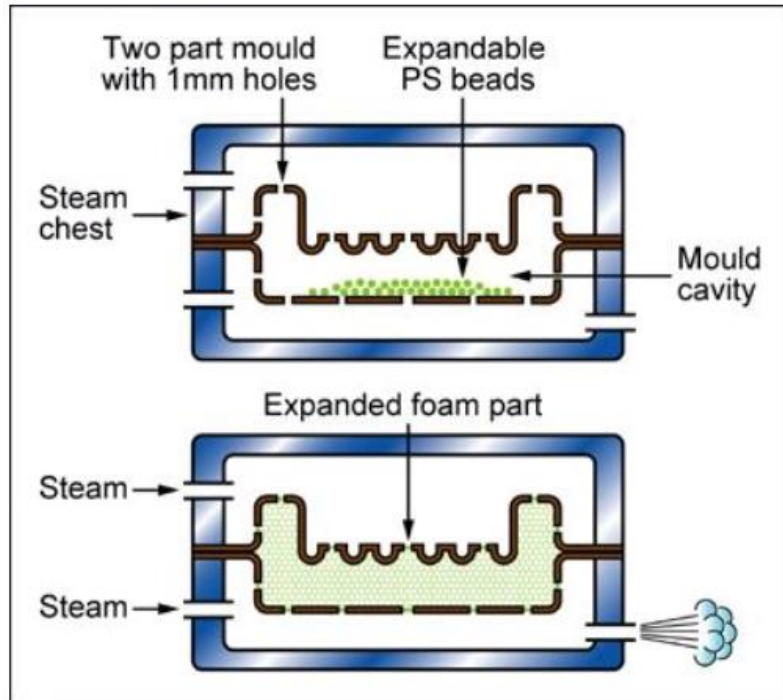


Expanded foam moulding

Two part mould with 1mm holes, expandable PS beads, steam chest, mould cavity, expanded foam part, steam, fart, mushroom cloud, green pellets, white cloud

Used to make foam surfboards

Process schematic



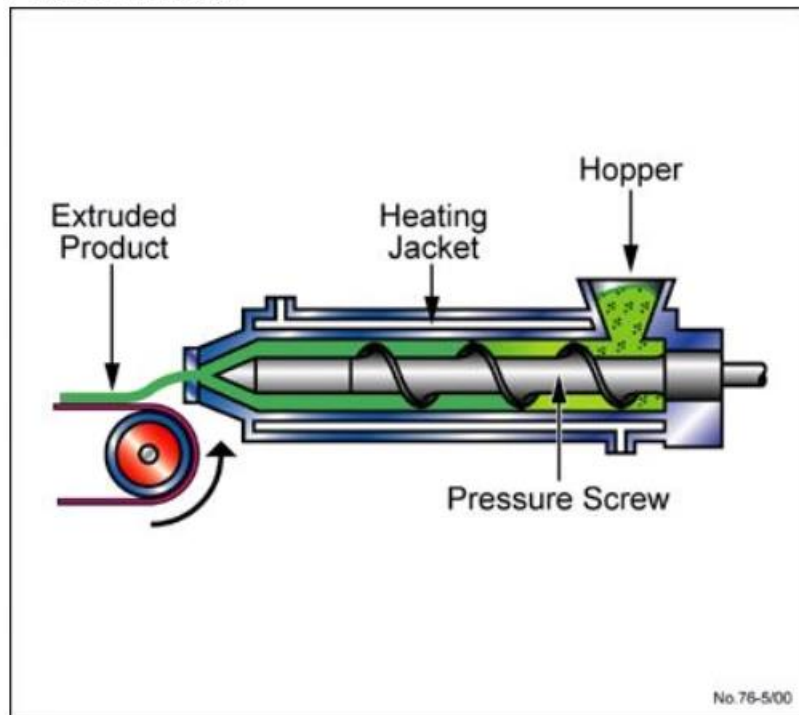
EMPTY SPACE

Like my brain in 501 exams fr

Polymer extrusion

Extruded product, heating jacket, hopper, pressure screw

Process schematic



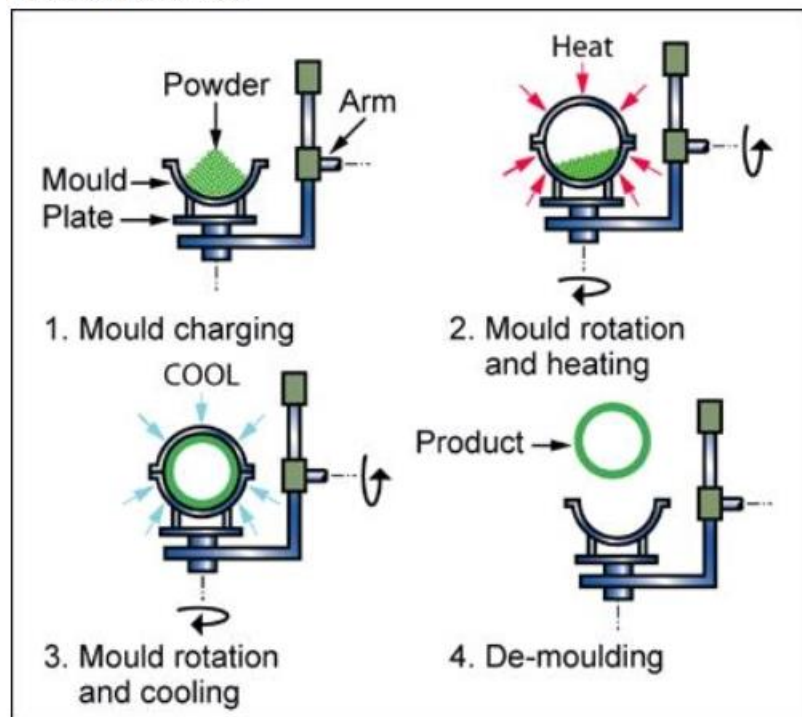
Rotational moulding

mould charging, mould rotation and heating, mould rotation and cooling, de-moulding

Powder, arm, mould plate, product, heat

Used to make plastic bins or something

Process schematic



Thermoforming

Vacuum forming

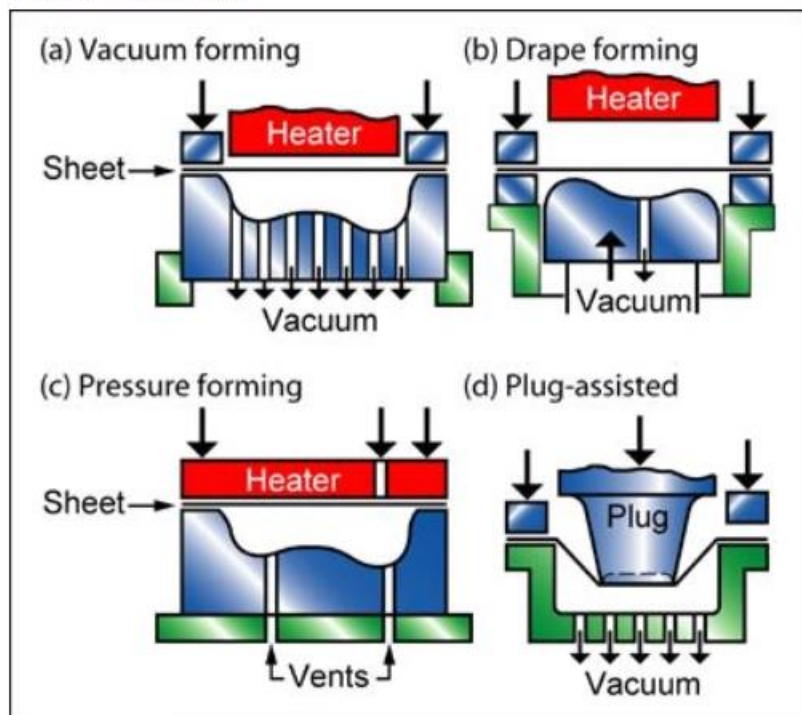
Drape forming

Pressure Forming

Plug-assisted forming

Heater, sheet, vacuum, vents, plug

Process schematic

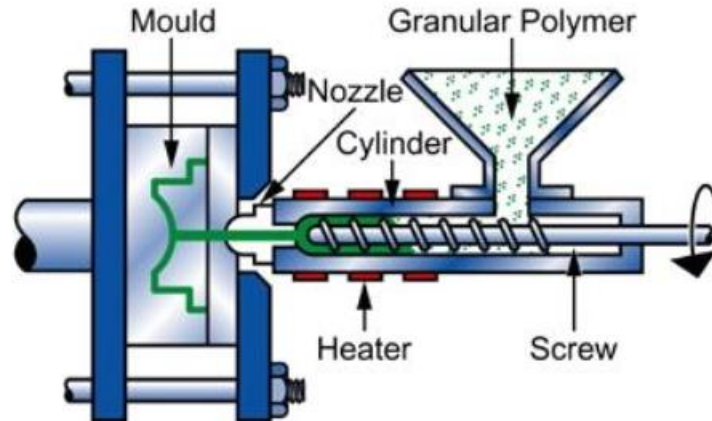


Injection moulding

Mould, nozzle, cylinder, granular polymer, cylinder, heater, screw

Used to make lego bricks

Process schematic



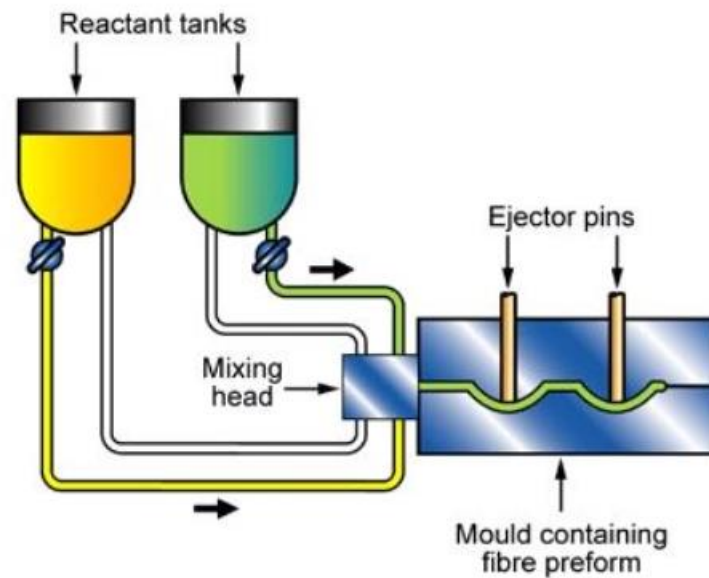
No.8-CMYK-5/01

Reaction injection moulding

Reactant tanks, mixing head, ejector pins, mould containing fibre preform

Note the little valves under the tanks, they are the only difference between this and smth similar. (RTM)

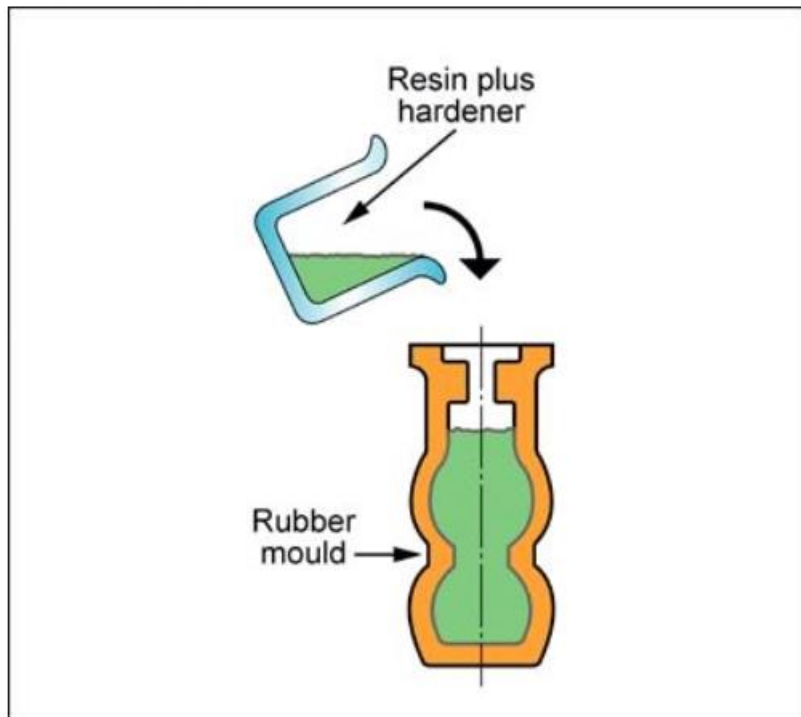
Process schematic



Resin casting

Resin plus hardener,
rubber mold

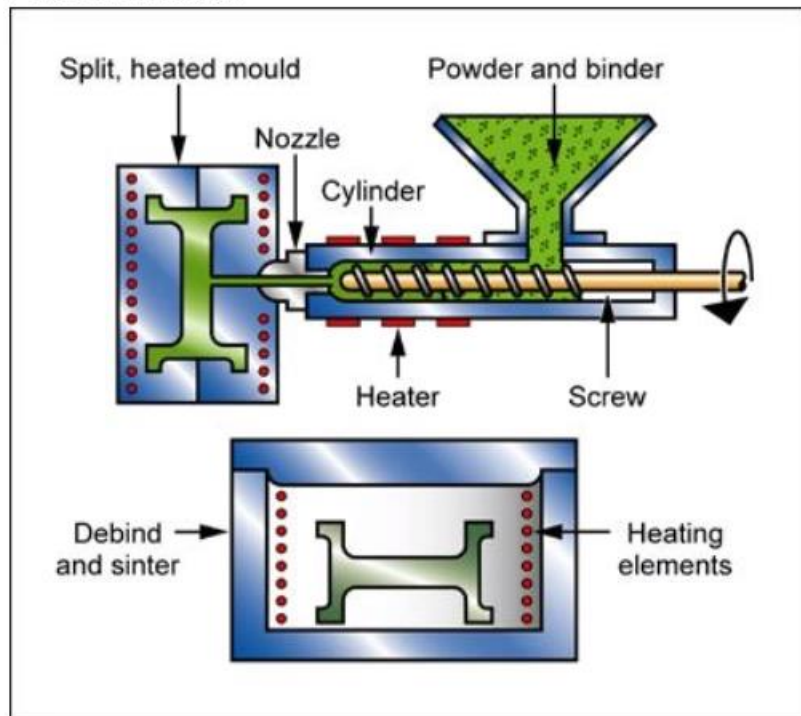
Process schematic



Ceramic (me thinks)
injection moulding

Split, heated mould,
nozzle, powder and binder,
cylinder, heater,
screw, debind and
sinter, heating elements

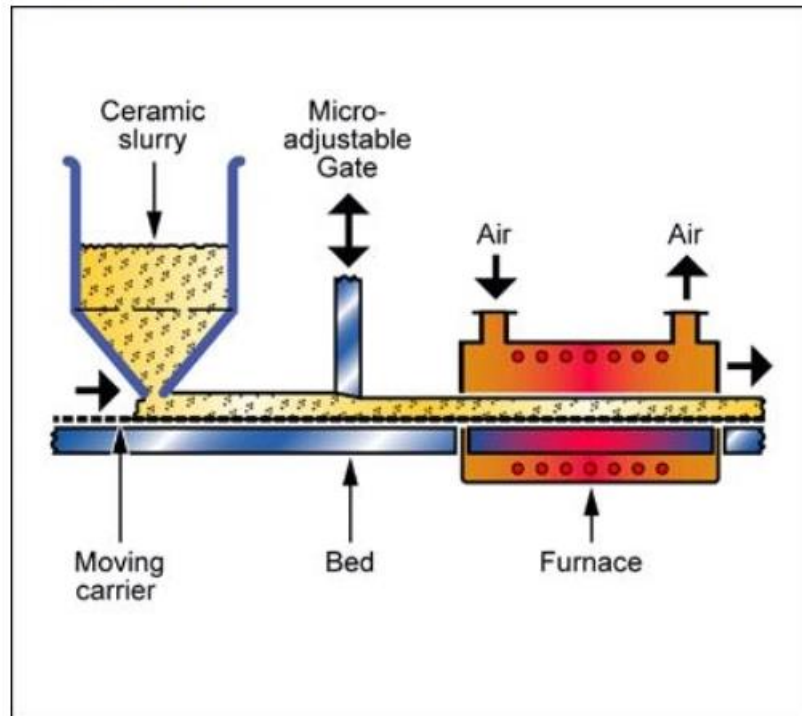
Process schematic



Tape casting

Ceramic slurry, moving carrier, bed, micro-adjustable gate, air, furnace, oven, roller, conveyer belt

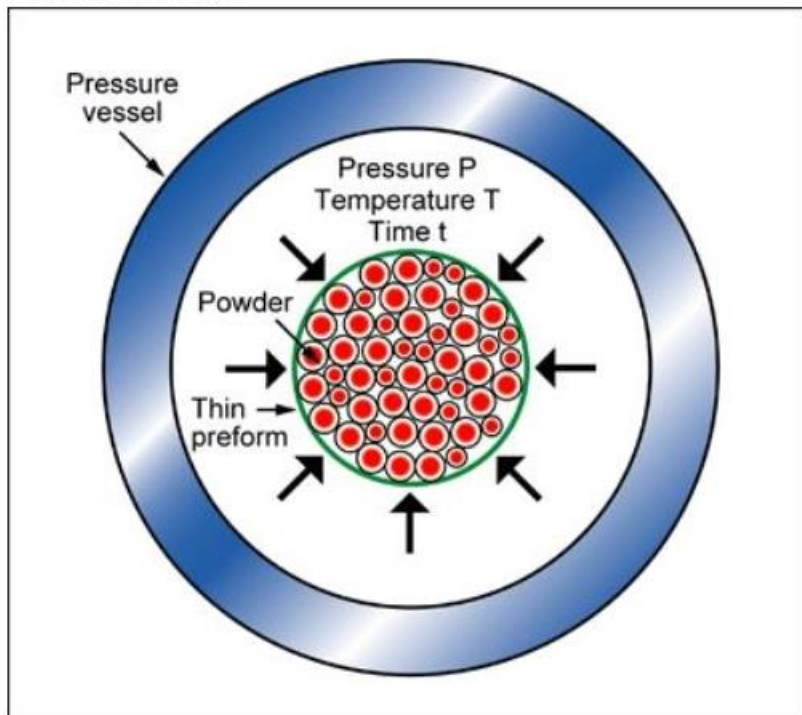
Process schematic



Hot isostatic pressing

Pressure vessel, circle, atoms, nucleus, powder, pressure P , temperature T , time t , Powder, thin preform

Process schematic

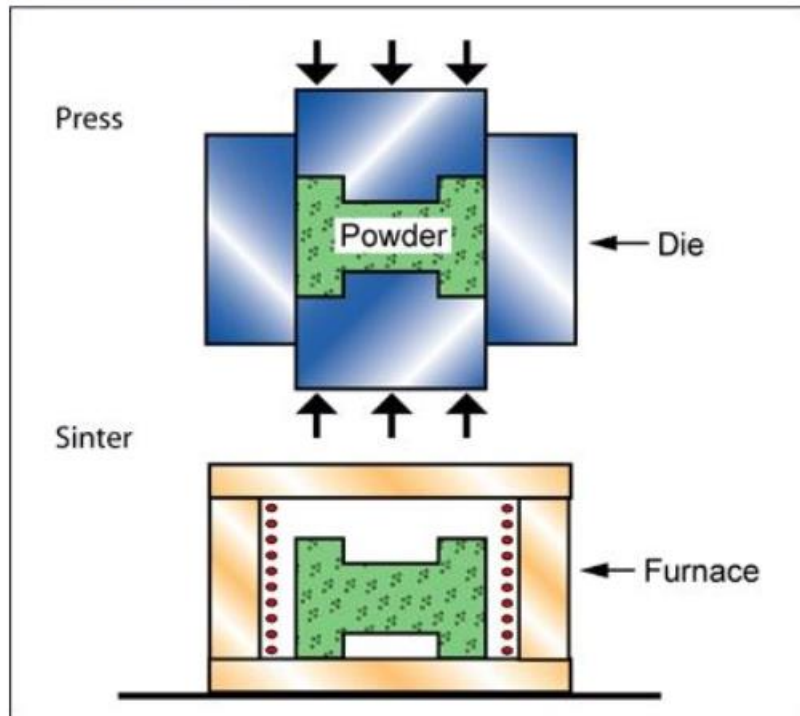


Pressing and sintering

Press, sinter

Powder, die, furnace

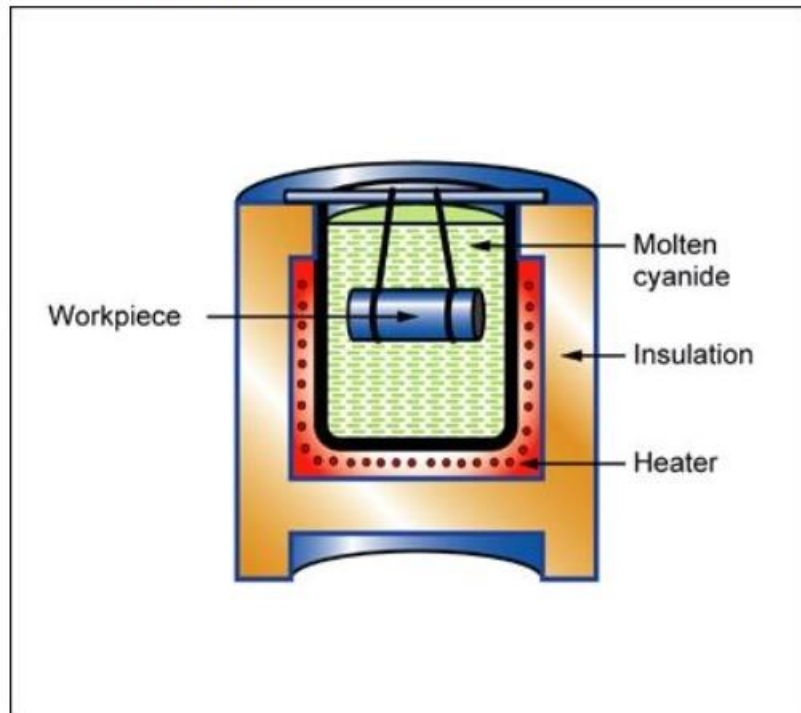
Process schematic



Heat treatment –
carburising and
carbonitriding

Workpiece, molten
cyanide, insulation,
heater

Process schematic

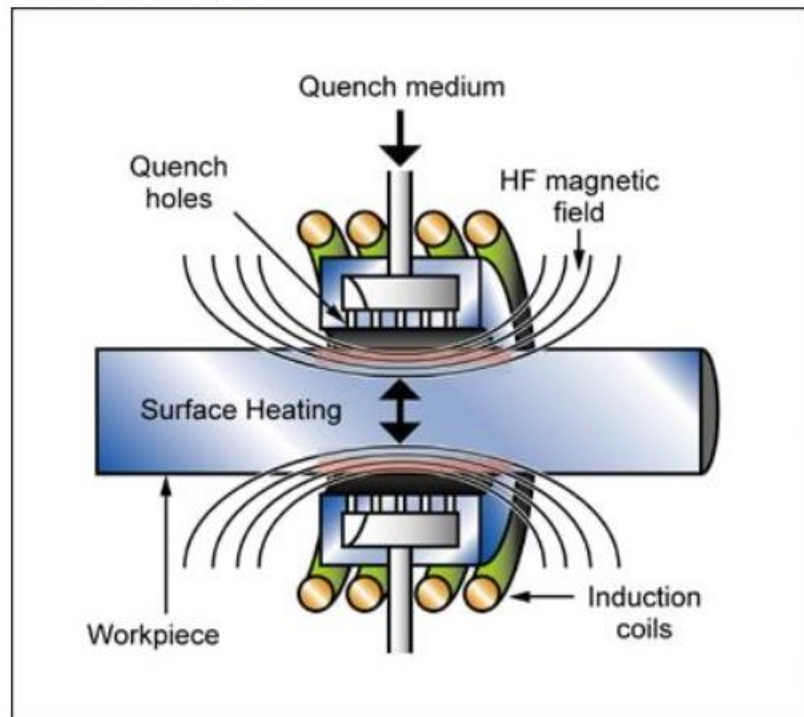


Induction and flame hardening

Quench holes, quench medium, HF magnetic field, surface heating, workpiece, induction coils

Hairs

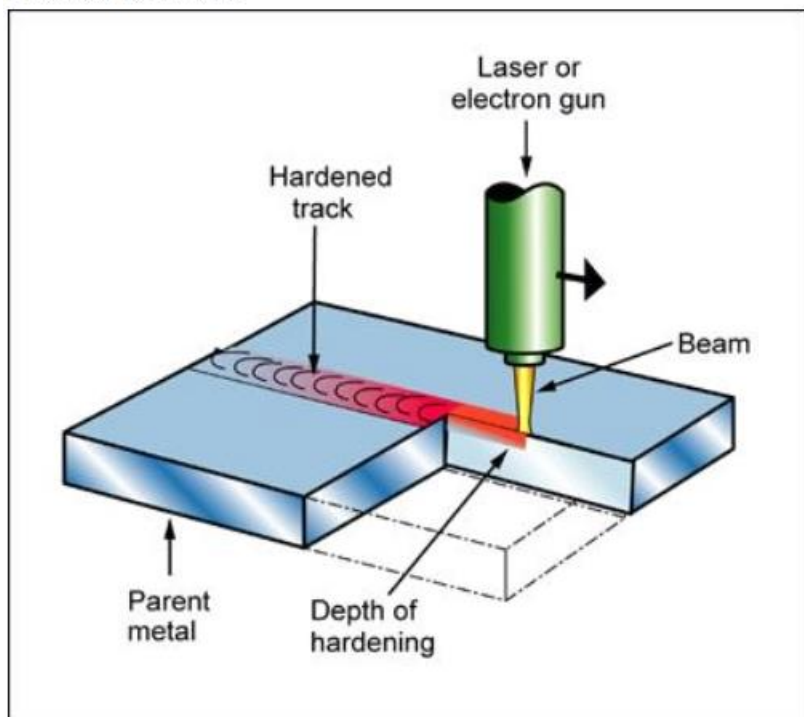
Process schematic



Laser hardening

Hardened track, laser or electron gun, beam, depth of hardening, parent metal

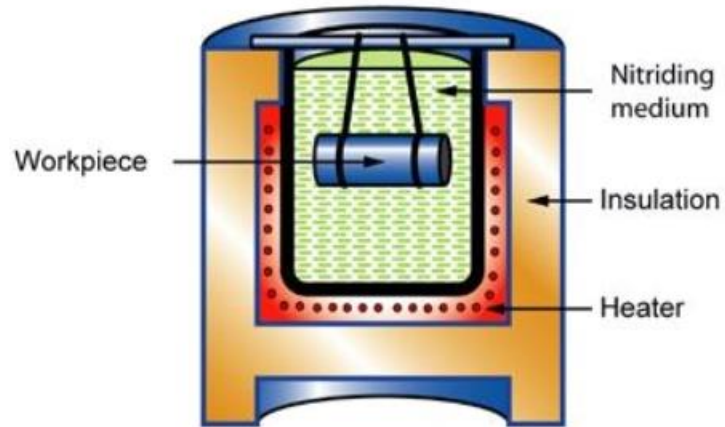
Process schematic



Nitriding

Workpiece, nitriding medium, insulation, heater

Process schematic

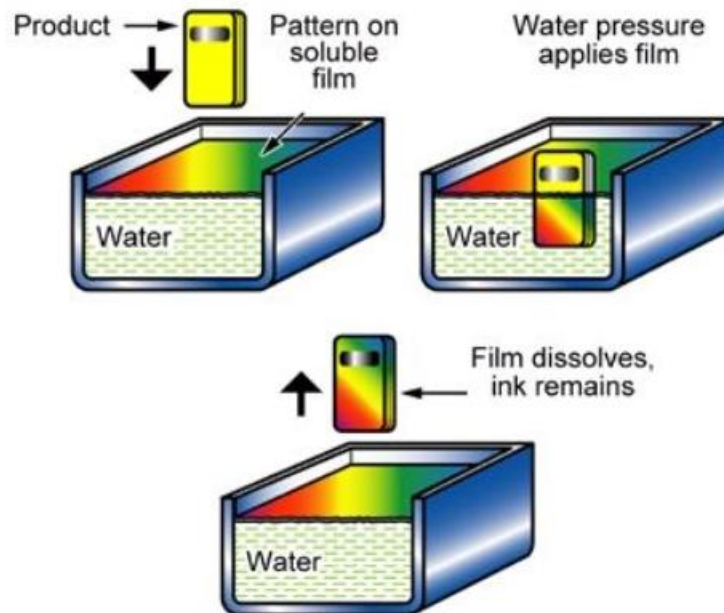


Painting and printing – cubic printing

Product, pattern on soluble film, water

Pattern on soluble film, water pressure applies film, film dissolves, ink remains.

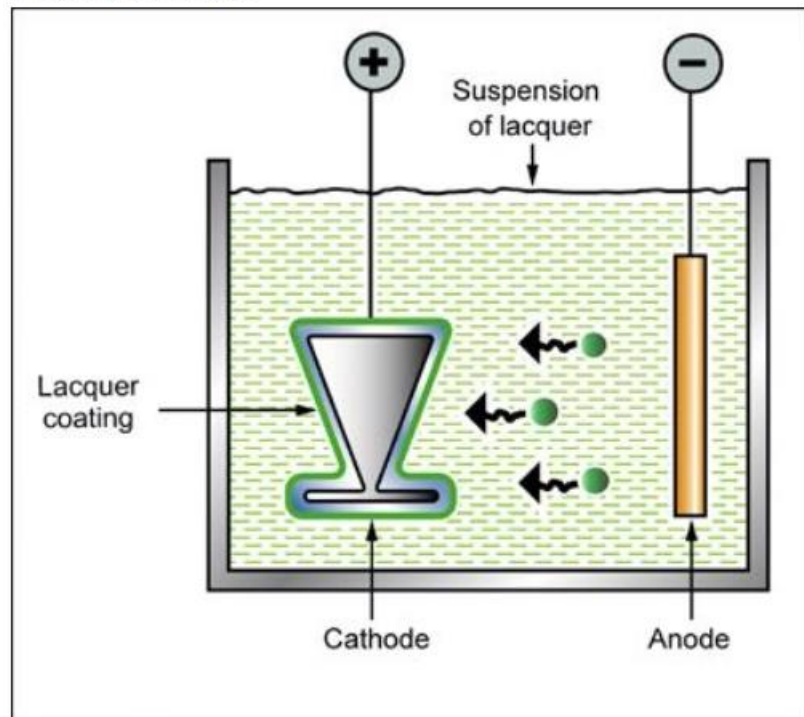
Process schematic



Electrophoretic and autophoretic
Lacquering (E-Coating)

Lacquer Coating,
Cathode, Anode,
Suspension of Lacquer

Process schematic

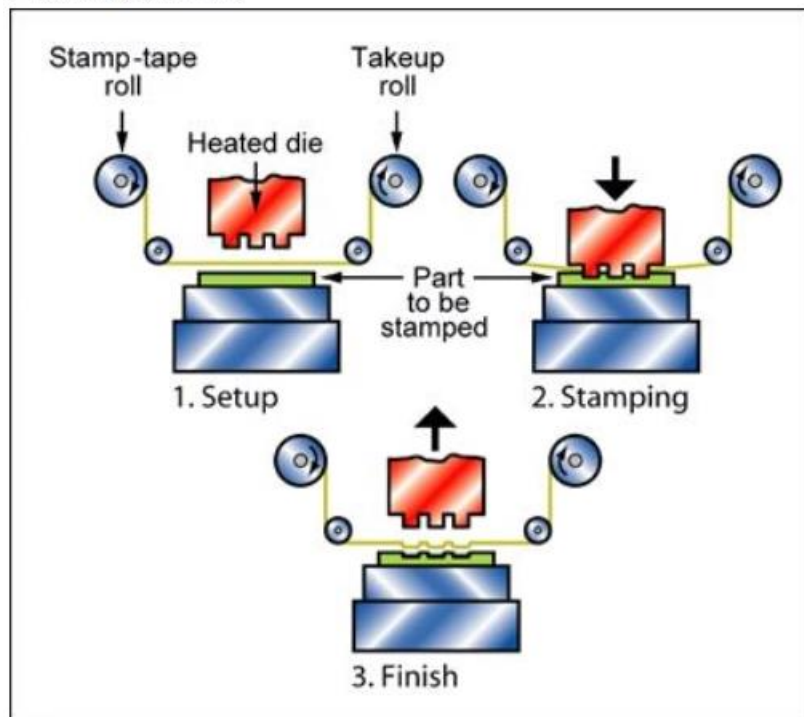


Hot Stamping

Setup, Stamping, Finish

Stamp-tape roll, heated die,
takeup roll, part to be stamped

Process schematic

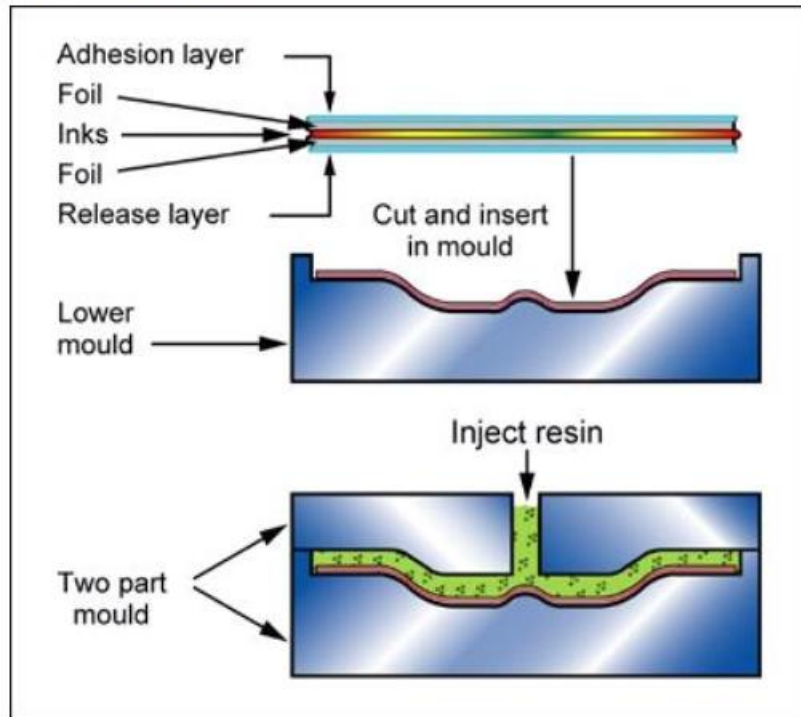


In-Mold Decoration (IMD)

Adhesion layer, foil, inks, release layer, lower mould, two part mould

Cut and insert in mould, inject resin

Process schematic

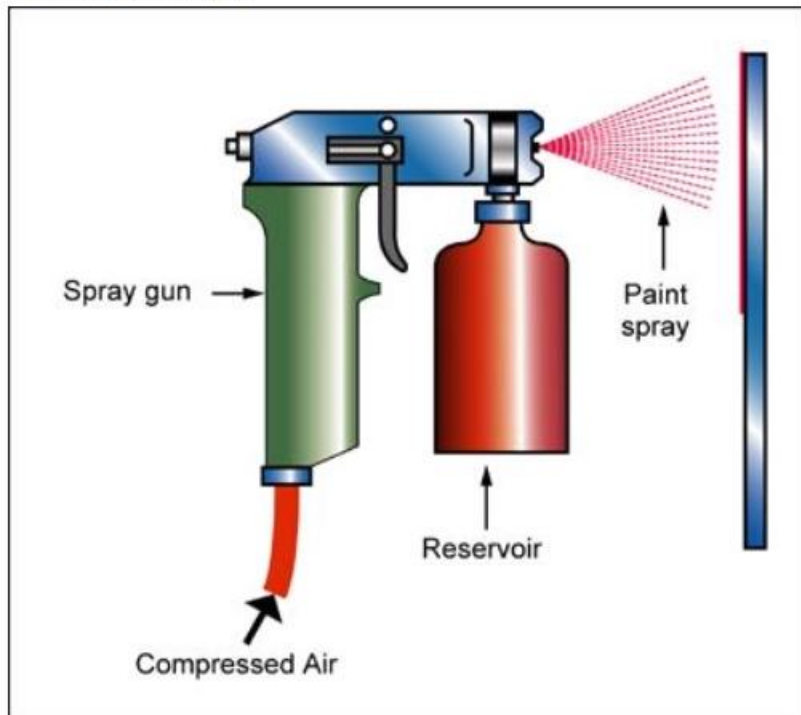


Spray Painting

Spray gun, paint spray, reservoir, compressed air

NOT PICTURED – Special version (Polymer Electrostatic spray) : very similar, but it sprays charged powder particles, and it's an electrostatic spray gun. Reservoir of powder rather than paint. Same photo, different labels.

Process schematic

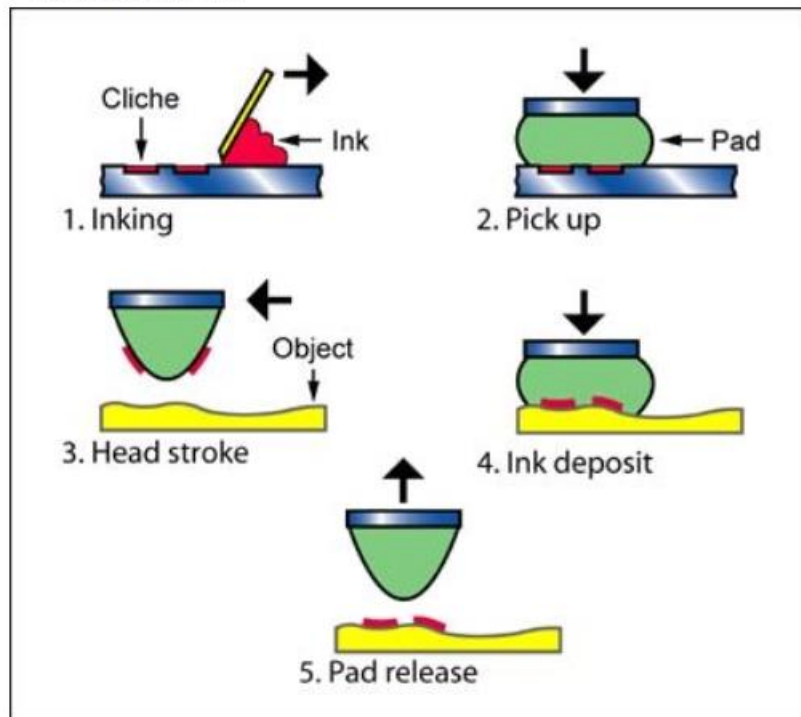


Pad Printing

Cliche, ink, pad, object

Inking, pick up, head stroke, ink deposit, pad release

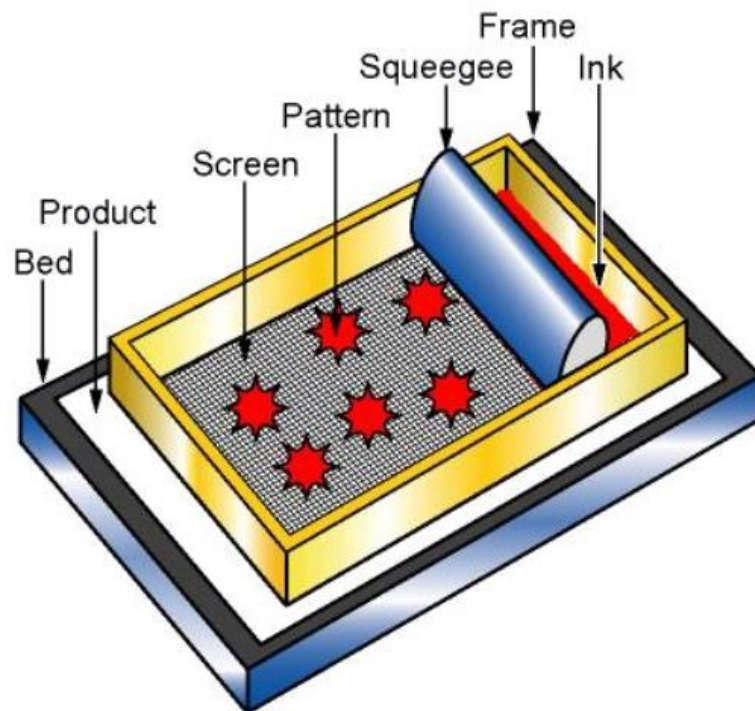
Process schematic



Silk Screen Printing

Bed, product, screen, pattern, squeegee, frame, ink

Process schematic

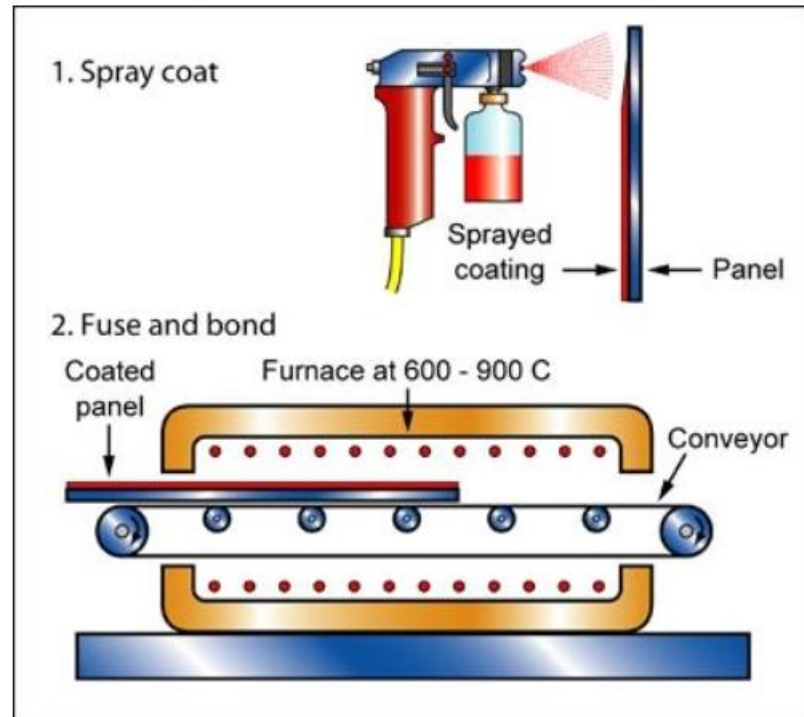


Vitreous Enameling

Spray coat, Fuse and bond

Sprayed coating, panel, Furnace at 600 – 900 C, conveyor

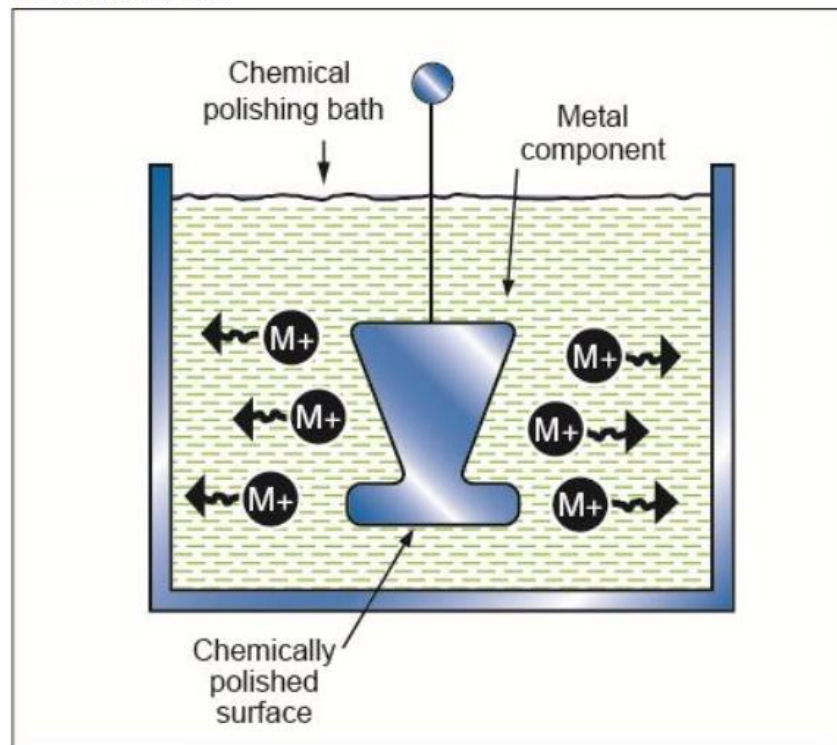
Process schematic



Chemical Polishing

Chemical polishing bath, metal component, chemically polished surface

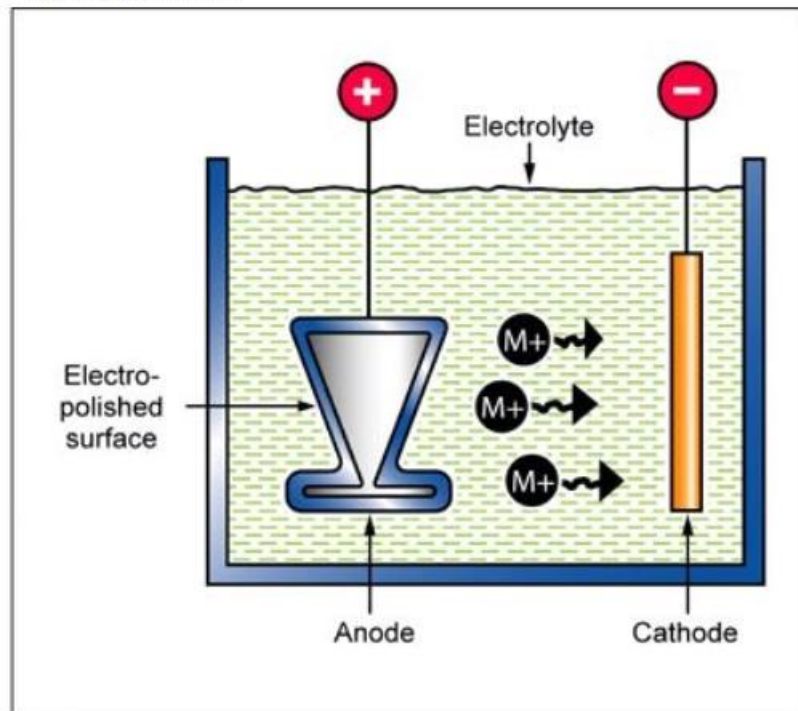
Process schematic



Electropolishing

Electro-polished surface, anode, cathode, electrolyte

Process schematic

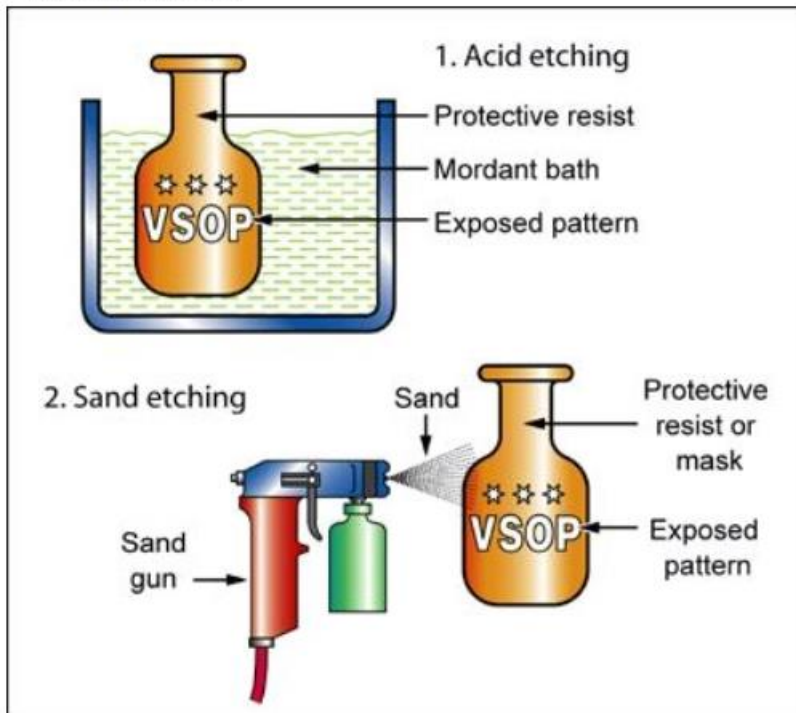


Etching

Acid Etching, Sand Etching

Protective resist, mordant bath, exposed pattern, sand gun, sand, protective resist or mask

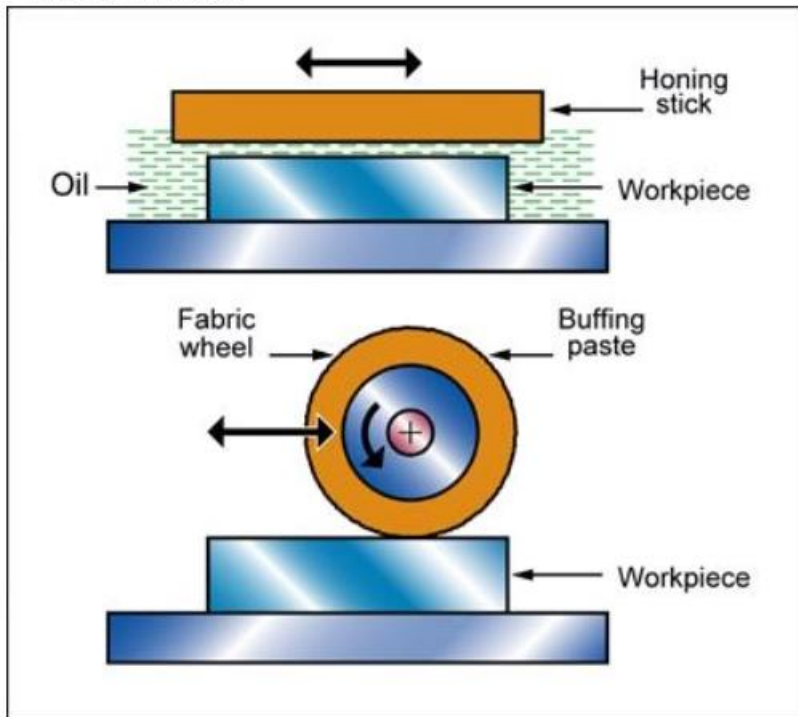
Process schematic



Micropolishing (Manual)

Oil, Honing Stick,
Workpiece, faric wheel,
buffing paste,
workpiece

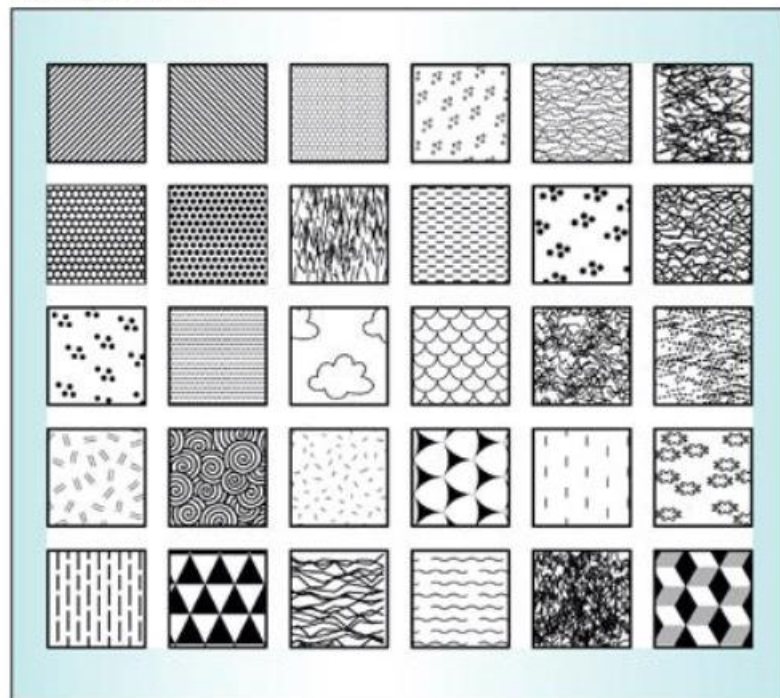
Process schematic



Texturing

Different patterns, cloth
pieces, threads, sewing,
texture, squares, art,
textiles , dress, curtains,
blankets, clouds ?

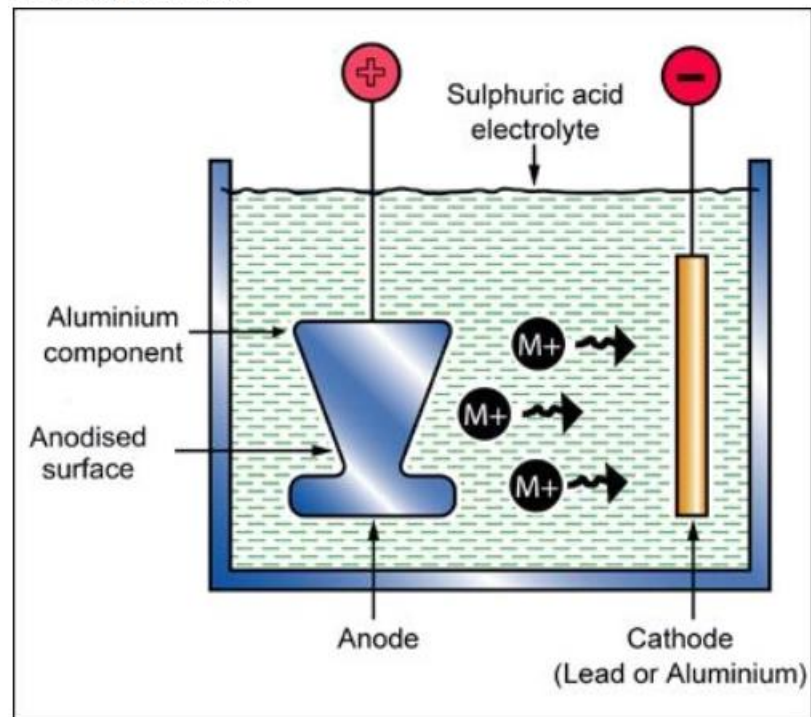
Process schematic



Anodizing / Anodising

Aluminium component,
anodized surface,
anode, cathode (lead or
aluminium), sulphuric
acid electrolyte

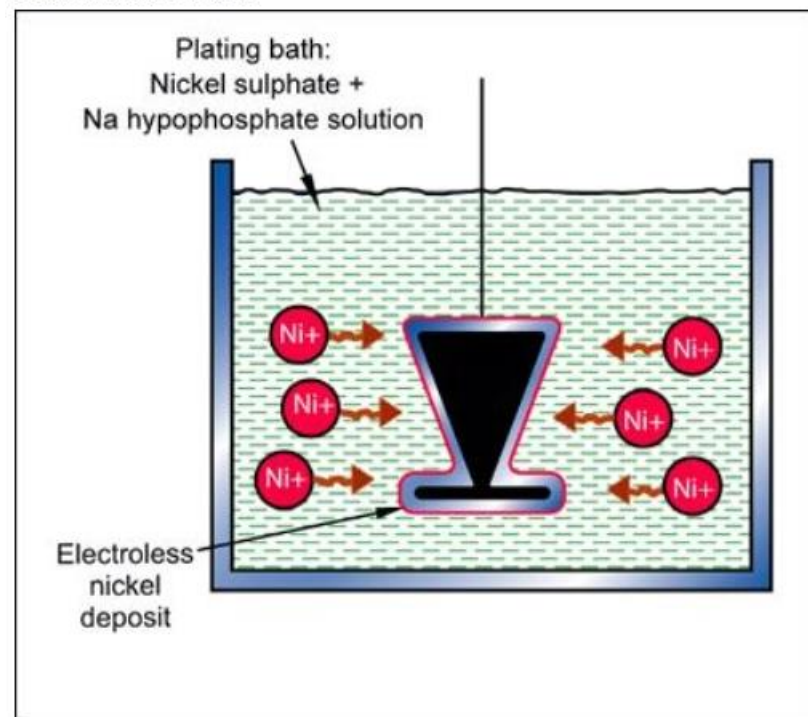
Process schematic



Chromating and phosphating

Plating bath : Nickel
sulphate + Na
hypophosphate
solution, Electroless
nickel deposit

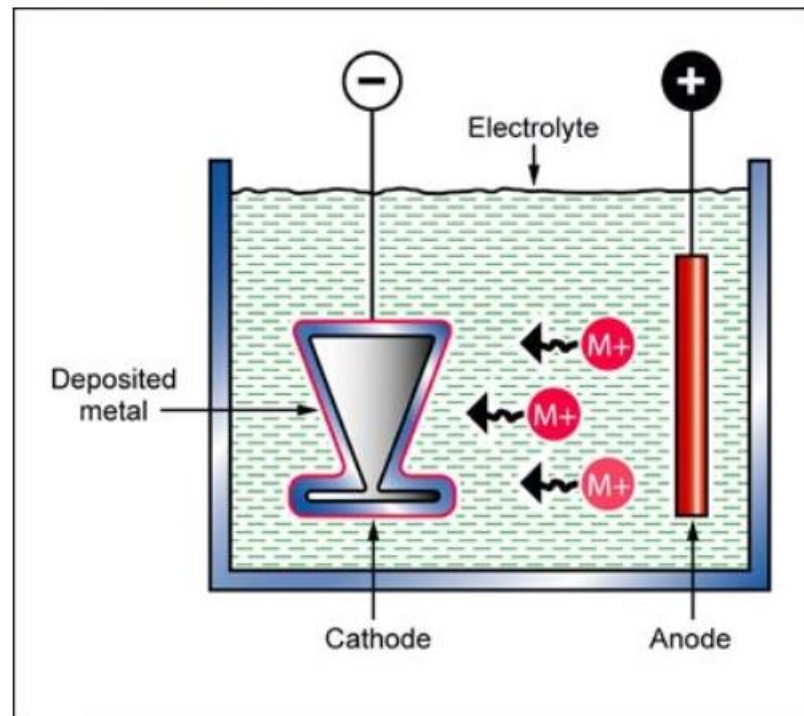
Process schematic



Electroplating

Deposited metal,
electrolyte, cathode,
anode

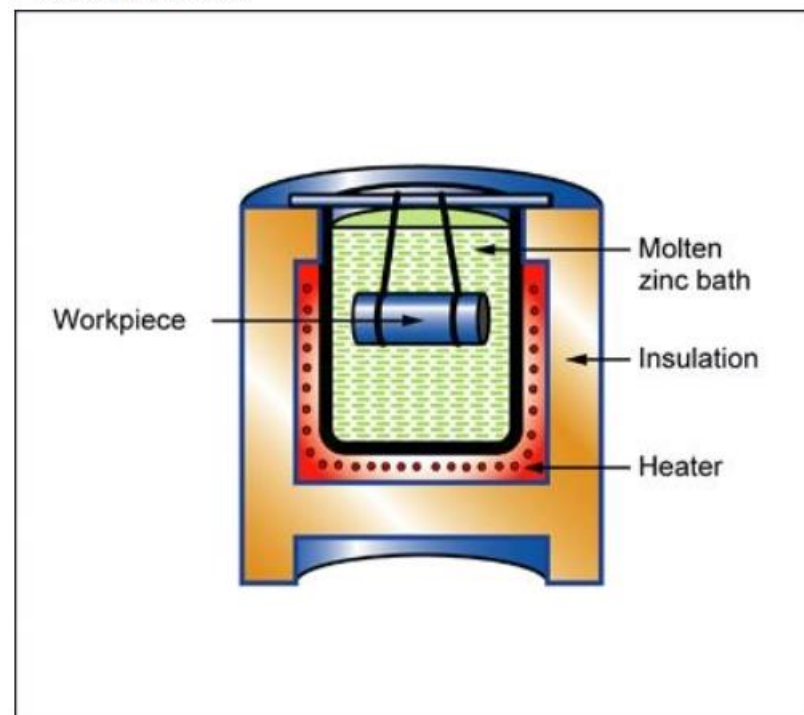
Process schematic



Galvanizing / galvenising

Workpiece, Molten zinc bath,
insulation, heater

Process schematic

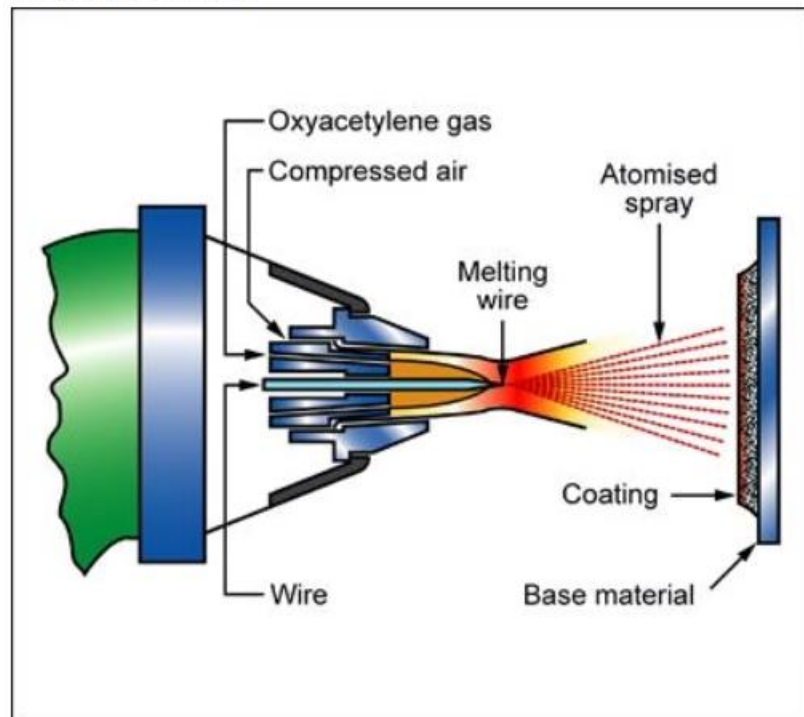


Thermal-spray Coating

Pen looking

Oxyacetylene gas,
compressed air, melting
wire, atomised spray,
coating, base material,
wire

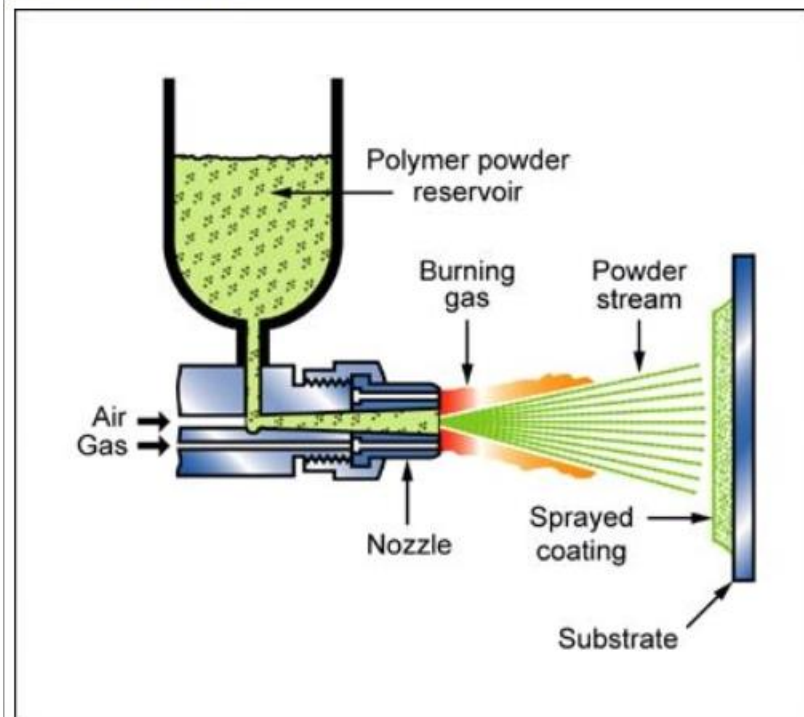
Process schematic



Polymer flame spraying / powder flame spraying

Polymer powder
reservoir, burning gas,
nozzle, air gas, powder
stream, sprayed
coating, substrate

Process schematic



PVD Plating

Component, deposit, evaporated atoms, heated evaporant, heater, vacuum.

Process schematic

